Gauss Law: MODULE-02 : GAUSS LAW.

The Electric flux ϕ passing through any closed surface is equal to total charge enclosed by the surface

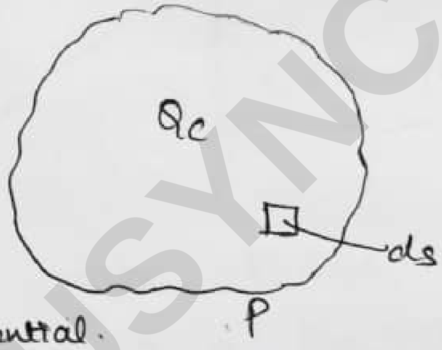
$$\therefore \psi_{total} = \oint D\, ds = Q_{enclosed}.$$

~~By~~ Gauss law is proved in two steps:

Step 1: $\psi_{total} = \oint D\, ds.$

Let Q Coulomb of charge is enclosed in a closed surface of any shape.

Let at a point P a differential element of area ds is assummed



The differential flux density coming out of the differential area is given by

$$d\psi = D \cos\theta \, ds.$$

Using dot product

$$d\psi = D \cdot ds \ c$$

The total flux ψ_{total} coming out of the closed surface is given by.

$$\psi_{total} = \oint D \cdot ds \ C.$$

• Step 2:

Let a point charge Q placed at the centre of imaginary Sphere of radius r.

The flux density due to a point charge is given by,

$$\boxed{D = \frac{Q}{4\pi r^2} a_r \ C/m^2} \rightarrow (1).$$

Since $\psi_{total} = \oint D.ds. \longrightarrow (2).$

$$\psi_{total} = \oint \frac{Q}{4\pi r^2} a_r . ds \longrightarrow (3).$$

for Spherical Co-ordinate

$$= \oint \frac{Q}{4\pi r^2} a_r r^2 \sin\theta \, d\theta \, d\phi \, dr.$$

$$= \frac{Q}{4\pi} \oint \sin\theta \, d\theta \, d\phi \, dr \, a_r.$$

$$= \frac{Q}{4\pi} \left[\int_{\theta=0}^{\pi} \sin\theta . d\theta \int_{\phi=0}^{2\pi} d\phi \right]$$

$$= \frac{Q}{4\pi} \left[-\cos[\pi] + \cos(0) \right] . [2\pi]$$

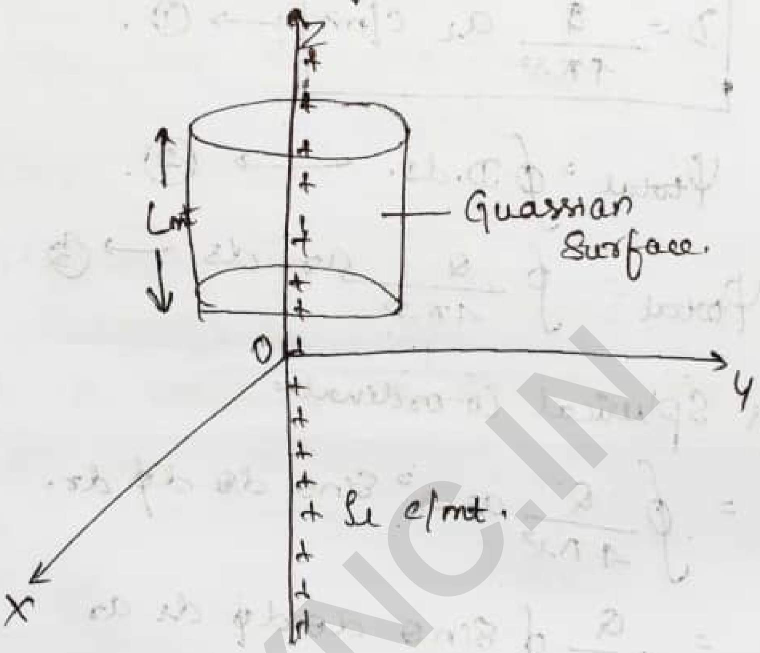
$$= \frac{Q}{4\pi} \left[-(-1) + 1 \right] 2\pi$$

$$= \frac{+\cancel{2}Q}{\cancel{4}\cancel{\pi}} \times \cancel{2}\cancel{\pi}$$

$$\boxed{\psi_{total} = +Q}$$

1. Application of Gauss law to Symmetrical Charge distribution.

The Gauss law is used to find electric field E and the flux density D due to line charges and sheet charges



The total charge enclosed by the length l is given as

$$Q = \rho_l \times L \longrightarrow (1)$$

the total flux ψ_{total} Coming out of Gaussian Surface

$$\psi_{total} = Q = \rho_l L \longrightarrow (2)$$

$$\psi_{total} = Q = \rho_l L = \oint D \cdot ds \longrightarrow (3).$$

$$\rho_l L = \oint D \cdot ds \longrightarrow (4)$$

$$\psi_{total} = \oint D\, ds \longrightarrow (5).$$

For a cylindrical co-ordinate system ds is given by,

$$ds = \rho\, d\rho\, d\phi\, dz$$

equation (5) can be re-written as

$$\Psi_{total} = \oint D_\rho\, \rho\, d\rho\, d\phi\, dz$$

$$= \oint D_\rho\, \rho\, dz\, d\phi\, d\rho$$

$$= \rho D_\rho \oint d\phi\, dz\, d\rho$$

$$= \rho D_\rho \left[\oint_0^{2\pi} d\phi \int_0^L dz\right]$$

$$\Psi_{total} = \rho D_\rho [2\pi] L = Q$$

$$\boxed{D_\rho = \frac{Q}{2\pi \rho L}} \longrightarrow (6)$$

Substitute (1) in (6)

$$D_\rho = \frac{\rho_L}{2\pi \rho}\ C/m^2$$

Electric field,

$$D = \varepsilon_0 E$$

$$E = \frac{D}{\varepsilon_0}$$

$$\boxed{E = \frac{\rho_L}{2\pi \rho\, \varepsilon_0}\ V/mt}$$

- **Numericals:**

1. Given the electric flux density $d = 0.3r^2\, a_r\, nc/m$ in free Space find

i) Electric field E at $P(r=2,\ \theta=25°,\ \phi=20°)$

ii) Q within the sphere of radius $r = 3$ mt.

iii) Total flux coming out of the sphere of radius $r = 4$ mtr.

Soln i) $D = \varepsilon_0 E$

$$E = \frac{D}{\varepsilon_0}$$

$$= \frac{0.3r^2\, a_r\ nc/m^2}{8.854 \times 10^{-12}}$$

$$= \frac{(0.3)(2\,mtr)^2\ a_r \times 10^{-9} /m^2}{8.854 \times 10^{-12}}$$

$$\boxed{E = 135.53\, a_r\ V/m}$$

ii) $Q = \oint D \cdot ds$

$$Q = \oint 0.3\, r^2\ a_r\ nc/m^2\ ds.$$

for Spherical Co-ordinate System

$$ds = r^2 \sin\theta\, d\theta\, d\phi\, dr$$

$$ds = r^2 \sin\theta\, d\theta\, d\phi\, a_r.$$

$$Q = \oint 0.3\, r^2\, a_r\, r^2 \sin\theta .\, d\theta .\, d\phi\, a_r$$

$$Q = 0.3\, r^4 \oint \sin\theta\, d\theta\, d\phi\ \underline{a_r \cdot a_r}^{1}$$

$$\underline{Q = 0.3\, r^4 \oint \sin\theta\, d\theta\, d\phi\ 2\pi}$$

$$Q = 0.3\,(3mt)^4 \left[\int_0^{\pi} \sin\theta\, d\theta \int_0^{2\pi} d\phi\right]$$

$= 0.3(81)\left[(-\cos\theta)_0^{\pi} \cdot (2\pi - 0)\right]$

$= (0.3)(81)\left[-\cos\pi + \cos(0) \quad [2\pi]\right]$

$= 24.3 \quad [(1)+(1)](2\pi)]$

$= 24.3 \times 4\pi$

$\boxed{Q = 305.36 nC}$

iii) $= \cancel{Q = 0.3}$

$\psi_{total} = Q = \oint D \cdot ds.$

$Q = 0.3\, r^4 \oint \sin\theta \, d\theta \, d\phi \, 2\pi$

$= (0.3)(4)^4 \int_0^{\pi} \sin\theta \, d\theta \int_0^{2\pi} d\phi.$

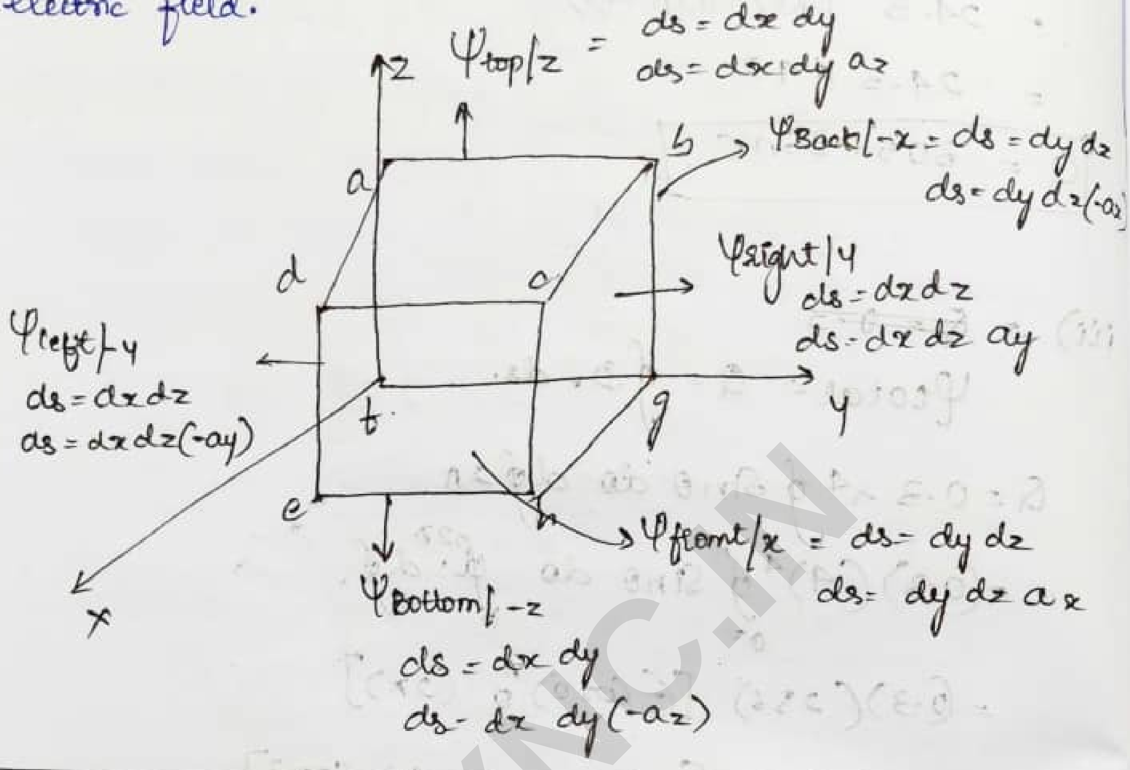
$= (0.3)(256) \left[(-\cos\theta)_0^{\pi} \, (2\pi)\right]$

$= (0.3)(256)\left[-\cos\pi + \cos(0) \, [2\pi]\right]$

$= (0.3)(\cancel{81}\,256)(4\pi)$

$\boxed{Q = 965.09 nC}$

ii) The Gauss law is used to find the total flux ψ_{total} and the charge enclosed by a cuboid [Closed Surface] when placed in non-uniform electric field.



→ Keynote:

i) $\psi_{top} = \psi_{Bottom} = 0$ $\qquad D_z = 0\ C/m^2$

$\psi_{top} + \psi_{Bottom} = 0$ $\qquad D_z \neq f(z)$

ii) $\psi_{front} = \psi_{Back} = 0$ $\qquad D_x = 0\ C/m^2$

$\psi_{front} + \psi_{Back} = 0$ $\qquad D_x \neq f(x)$

iii) $\psi_{right} = \psi_{left} = 0$ $\qquad D_y = 0\ C/m^2$

$\psi_{right} + \psi_{left} = 0$ $\qquad D_y \neq f(y)$

1. A Cube of 4 mtrs. Centered at the Origin with edges parallel to the Co-ordinate axis of Cartesian Co-ordinate System if flux density $D = \frac{20x^2}{5} a_x$ c/m². What is the total charge Contained in the Q.

Sol $D_x = 4x^2 a_x$ c/m²
$D_y = 0$
$D_z = 0$

as $D_y = 0$
$\psi_{right} = \psi_{left} = 0.$

as $D_z = 0$
$\psi_{top} = \psi_{Bottom} = 0.$

$\psi_{total} = \psi_{front} + \psi_{Back}.$

$\psi_{front} = \oint D \cdot ds$

$= \oint 4x^2 a_x \; dy \, dz \; a_x$

$= \oint 4x^2 \, dy \, dz$

$= 4x^2 \oint_{-2}^{2} dy \oint_{-2}^{2} dz$

$= 4x^2 [2+2][2+2]$

$\boxed{\psi_{front} = 64x^2}$

$x = 2$

$\psi_{front} = 64(4)$

$\boxed{\psi_{front} = 256 \text{ nc}}$

$$\psi_{Back} = \oint D \cdot ds$$
$$= \oint 4x^2 a_x \cdot dy\, dz\, (-a_x)$$
$$= 4x^2 \oint dy \oint dz\, (a_x)(-a_x)$$
$$= -4x^2 \oint_{-2}^{2} dy \oint_{-2}^{2} dz.$$
$$= -4x^2 [y]_{-2}^{2} [z]_{-2}^{2}$$
$$= -4x^2 [2+2] [2+2]$$
$$= (-4)(2)^2 (4)(4).$$
$$\boxed{\psi_{Back} = -256\ nC}$$

$$\psi_{total} = \psi_{front} + \psi_{Back}$$
$$= 256 - 256$$
$$\boxed{\psi_{total} = 0}$$

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2. A Cube of 6 mtr Centered at Origin with edges parallel to co-ordinates axis of Cartesian Co-ordinate System. If flux density $D = 6y^5 a_y$ C/m², what is the total flux Coming Out of the Cube.

Sol $D_y = 6y^5 a_y$ C/m²

as $D_x = 0$ $D_z = 0$

$$\psi_{total} = \psi_{right} + \psi_{left}.$$

$$\psi_{right} = \oint D \cdot ds$$
$$= \oint 6y^5 a_y\, dx\, dz\, a_y.$$

$$= 6y^5 \oint dx \oint dz$$

$$= 6y^5 \oint_{-3}^{3} dx \int_{-3}^{3} dz$$

$$= 6(3)^5 [3+3] [3+3]$$

$$= (6)(243)(6)(6)$$

$$\boxed{\psi_{right} = 52488}$$

$$\psi_{left} = \oint D \cdot ds$$

$$= \oint 6y^5 a_y \, dx \, dz \, (-a_y)$$

$$= -6y^5 \oint_{-3}^{3} dx \int_{-3}^{3} dz$$

$$= -(6)(3)^5 (3+3)(3+3)$$

$$\boxed{\psi_{left} = -52488}$$

$$\psi_{total} = \psi_{\cancel{front}\,right} + \psi_{left}$$

$$= 52488 - 52488$$

$$\boxed{\psi_{total} = 0}$$

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3. Find the total charge in a volume defined by 6 planes for which $1 \leq x \leq 2$, $2 \leq y \leq 3$, $3 \leq z \leq 4$ if $D = 4x\,a_x + 3y^2 a_y + 2z^2 a_z$ c/m^2.

Sol $\psi_{top} = \oint D_z ds$

$$= \cancel{\oint (4x a_x + 3y^2 a_y + 2z^2 a_z)}$$

$$= \oint (2z^2 a_z) \, dx \cdot dy \, a_z$$

$$\Psi_{top} = \oint 2z^2 \, dx \, dy$$ (limits 2, 3)

$$= 2z^2 \oint_1^2 [dx] \oint_2^3 dy$$

$$= 2z^2 [x]_1^2 \, [y]_2^3$$

$$= 2z^2 [2-1] [3-2]$$

$$= 2z^2$$

where $z = 4$

$$= 2(4)^2$$

$$\boxed{\Psi_{top} = 32C}$$

$$\Psi_{Bottom} = \oint D \, ds$$

$$= \oint 2z\hat{a}_z \, dx \, dy \, (-a_z)$$

$$= \oint -2 z^2 \, dx \, dy$$

$$= -2z^2 [x]_1^2 \, [y]_2^3$$

$$= -2(3)^2 (1)(1)$$

$$\boxed{\Psi_{Bottom} = -18C}$$

$$\Psi_{right} = \oint D \, ds$$

$$= \oint 3y^2 a_y \, dx \, dz \, (a_y)$$

$$= 3y^2 \oint_1^2 dx \cdot \oint_3^4 dz$$

$$= 3y^2 [2-1] \cdot [4-3]$$

$$= 3(3)^2$$

$$\boxed{\Psi_{right} = 27C}$$

$$\psi_{left} = \oint D \cdot ds$$

$$= \oint 3y^2 \, a_y \;\; dx \;\; dy(-a_y)$$

$$= -3y^2 \oint_1^2 dx \oint_3^4 dz$$

$$= -3y^2 [2-1] [4-3]$$

$$= -3 (2)^2$$

$$\boxed{\psi_{left} = -12C}$$

$$\psi_{front} = \oint D \cdot ds$$

$$= \oint 4x \, a_x \cdot \; dy \, dz \, a_x$$

$$= 4x \oint_2^3 dy \oint_3^4 dz$$

$$= 4x [3-2] [4-3]$$

$$= 4(2)$$

$$\boxed{\psi_{front} = 8C}$$

$$\psi_{Back} = \oint D \cdot ds$$

$$= \oint 4x \, a_z \;\; dy \;\; dz (-a_x)$$

$$= -4x \oint_2^3 dy \oint_3^4 dz$$

$$= -4(1)\;(1)(1)$$

$$\boxed{\psi_{Back} = -4C}$$

$$\psi_{total} = \psi_{front} + \psi_{Back} + \psi_{left} + \psi_{Right} + \psi_{top} + \psi_{Bottom}$$

$$= 32C - 18C + 24C - 12C + 4C + 8C$$

$$\boxed{\psi_{total} = 33C = Q}$$

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4. Given in a Certain Region of Space $D = 2xy\, a_x + 3yz\, a_y + 42x\, a_z$ C/m². Evaluate the amount of flux that passes through the portion bounded by $-1 \le y \le 2$ and $0 \le z \le 4$ in $x = 3$ plane.

Sol

$$\psi_{front} = \oint D_x\, ds$$

$$= \oint 2xy\, a_x\, dy\, dz\, a_x$$

$$= \oint 2xy\, dy\, dz$$

$$= 2xy \oint dy \oint dz$$

$$= 2x\cancel{y} \oint_{-1}^{2} y\,dy \oint_{0}^{4} dz$$

$$\cancel{= 2x\left[\frac{y^2}{2}\right] [y]_{-1}^{2} [z]_0^4}$$

$$\cancel{= 2x[y]^2 [2+1] [4]}$$

$$\cancel{= (2)(3)(2)^2(3)(4)}$$

$$\cancel{= 144C}$$

$$= 2x\left[\frac{y^2}{2}\right]_{-1}^{2} [z]_0^4$$

$$= \frac{\cancel{2}x}{\cancel{2}} [2^2 - (1)^2] [4]$$

$$= x[3][4]$$

$= (3)(3)(4)$

$\boxed{\psi_{front} = 36\,C}$

$\boxed{\psi_{front} = \psi_{total} = 36\,C}$

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5. Given $60\mu C$ point charge located at origin, find the total flux passing through
i) Portion of the sphere of the radius $r = 26mt$ bounded by $0 < \theta < \frac{\pi}{2}$, $0 < \phi < \frac{\pi}{2}$.
ii) Closed surface defined by $\rho = 26mt$, ~~z = ±26mt~~.
iii) The plane $z = 26mt$

Sol: The flux density due to a point charge is given by,

i) $$D = \frac{Q}{4\pi r^2} a_r \; c/m^2$$

$$\psi = \oint \frac{Q}{4\pi r^2} a_r \, ds$$

$ds = r^2 \sin\theta \, d\theta \, d\phi \, dr$

$ds = r^2 \sin\theta \, d\theta \, d\phi \, a_r$

$$\psi = \frac{Q}{4\pi} \oint \frac{1}{\cancel{r^2}} a_r \cancel{r^2} \sin\theta \, d\theta \, d\phi \, a_r$$

$$= \frac{Q}{4\pi} \int_0^{\pi/2} \sin\theta \, d\theta \, d\phi$$

$$= \frac{Q}{4\pi} \left[-\cos\theta\right]_0^{\pi/2} \left[\phi\right]_0^{\pi/2}$$

$$= \frac{Q}{4\pi} \left[-\cos\frac{\pi}{2} + \cos(0)\right] \left[\frac{\pi}{2}\right] = \frac{60\times10^{-6}}{4\pi} \left[0+1\right] \left[\frac{\pi}{2}\right]$$

$$\boxed{\psi = 7.5\,\mu C}$$

ii) $\rho = 26mt$

$$\psi = \oint D \cdot ds$$

$$= \oint \frac{Q}{4\pi\rho} a_\rho \, ds.$$

$$= \oint \frac{Q}{4\pi\rho} \cancel{a_\rho} \, \cancel{d\rho} \, d\phi \, dz$$

$$= \oint \frac{Q}{4\pi\rho} \int_0^{2\pi} d\phi \int_{-26}^{26} dz$$

$$= \oint \frac{60 \times 10^{-6}}{4 \times \pi \times 26} [\phi]_0^{2\pi} [z]_{-26}^{26}$$

$$= \frac{60 \times 10^{-6}}{4 \times \pi \times 26} [2\pi] [26 + 26]$$

$$\boxed{\psi_{total} = 60 \mu C}$$

iii) $z = 26 \, mt.$

$$\psi = \oint D \cdot ds$$

$$= \oint \frac{Q}{4\pi\rho} a_\rho \, d\rho \, d\phi \, dz$$

$$= \frac{60 \times 10^{-6}}{4\pi \times 26} \oint_0^{2\pi} d\phi \oint_0^{26} dz$$

$$= \frac{60 \times 10^{-6}}{4\pi \times 26} [2\pi] [26]$$

$$\boxed{\psi_{total} = 30 \mu C}$$

6. A Given 5nC point charge is located at the origin. Find the flux passing through the following Surfaces.

i) $\rho = 10$ mtrs $\quad 0 \le \phi < \pi$, $z = 10$ mtrs.

ii) $r = 10$ mtrs $;\ 0 < \theta < \pi$, $\quad 0 < \phi < \pi$

S) i) $\psi = \oint D \cdot ds$

$$= \oint \frac{Q}{4\pi\rho} \cancel{a_\rho}\, d\phi\, dz\, \cancel{a_\rho}$$

$$= \frac{Q}{4\pi\rho} \oint_0^{\pi} d\phi \oint_0^{10} dz$$

$$= \frac{\cancel{60} \times 5 \times 10^{-9}}{4\pi \times 10} \times [\pi][10]$$

$$\boxed{\psi_{total} = 1.25\, nC}$$

ii) $\psi = \oint D \cdot ds$

$$= \oint \frac{Q}{4\pi \cancel{r^2}} \cancel{d\theta\, d\phi}\ \cancel{r^2} \sin\theta\, d\theta\, d\phi\, \cancel{a_r}$$

$$= \frac{5 \times 10^{-9}}{4\pi \times \cancel{10}} \oint_0^{\pi} \frac{d\theta}{\sin\theta} \oint_0^{\pi} d\phi$$

$$= \frac{5 \times 10^{-9}}{4\pi \times \cancel{100}} [\pi]\ [\cancel{\pi}]\ [-\cos\theta]_0^{\pi}$$

$$= \frac{5 \times 10^{-9}}{4\cancel{\pi}} (\cancel{\pi}) (-\cos\pi + \cos 0)$$

$$= \frac{5n}{\cancel{4}_2} (\cancel{2})$$

$$\boxed{\psi = 2.5\, nC}$$

7. Find the total charge in a volume defined by 6 planes for which $1 \le z < 2$, $2 < y < 3$, $3 < x < 4$ if $D = (2y^2z - 8xy)a_x + (4xyz - 4x^2)a_y + (2xy^2 - 4z)a_z$

Sol $\psi_{front} = \oint D_x \, ds$

$= \oint (2y^2z - 8xy)\, \cancel{a_x}\, dy\, dz\, \cancel{a_x}$

$= \oint \cancel{2y^2z - 8xy}\, \cancel{dy\, dz}\, \cancel{dz}$

$= \oint 2y^2z\, dy\, dz - \oint 8xy\, dy\, dz$

$= 2\left[\frac{y^3}{3}\right]_2^3 \left[\frac{z^2}{2}\right]_1^2 - 8x\left(\frac{y^2}{2}\right)_2^3 [dz]_1^2$

$= \frac{2}{6}[3^3 - 2^3][2^2 - 1^2] - \frac{8(4)}{2}[3^2 - 2^2](2-1)$

$\boxed{\psi_{front} = -61C}$

$\psi_{Back} = \oint 2y^2z\, dy\, dz + \oint 8xy\, dy\, dz$ $(-a_x)$ $\int(1)$

$= 2\int_2^3 y^2 dy \int_1^2 z\, dz + 8x\int_2^3 y\, dy \int_1^2 dz$

$= 2\left(\frac{19}{3}\right)\left(\frac{3}{2}\right) + 8x\left(\frac{5}{2}\right)(1)$

$x = 3$

$\boxed{\psi_{Back} = 79C}$

$\psi_{top} = \oint (2xy^2 - 4z)\, a_z\, dy\, dx\, a_z$

$= \int 2xy^2 dy\, dx - 4\int z\, dy\, dx$

$= 2\int_3^4 x\, dx \int_2^3 y^2 dy - 4z\int_2^3 dy \int_3^4 dx$

$= \cancel{2}\frac{[4^2 - 3^2]}{\cancel{2}} \frac{[3^3 - 2^3]}{3} - 4z(3-2)(4-3)$

$$= (16-9)\left(\frac{27-8}{3}\right) - 4z(1)(1) \qquad z=2$$

$$\boxed{\Psi_{top} = 36.33\ C}$$

$$\Psi_{Bottom} = \int 2xy^2\, a_z a_z\, dy\, dx - 4\int z\, dy\, dx\, a_z(-a_y)$$

$$= 2\int_3^4 x\, dx \int_2^3 y^2\, dy + 4z\int_0^3 dy \int_3^4 dx$$

$$= 2\frac{[4^2-3^2]}{2}\frac{[3^2-2^2)}{3} + 4z(3-2)(4-3)$$

$$= (6-9)\left(\frac{27-8}{3}\right) + 4z \qquad z=1$$

$$\boxed{\Psi_{Bottom} = 48.33C}$$

$$\Psi_{right} = \int (4xyz - 4x)\, a_y\ a_y\, dx\, dz$$

$$= 4y\int_3^4 x\, dx \int_1^2 z\, dz - 4\int_3^4 x^2 dx \int_1^2 dz$$

$$= 4y\left(\frac{1}{1}\right)\left(\frac{3}{2}\right) - 4\left(\frac{37}{3}\right)(1)$$

$$= 21y - \frac{148}{3} \qquad y=3$$

$$\boxed{\Psi_{right} = 13.67C}$$

$$\Psi_{left} = \int 4xyz\, dx\, dz + 4\int_3^4 x^2 dx \int_1^2 dz$$

$$= 21y + \frac{148}{3} \qquad y=2$$

$$\boxed{\Psi_{left} = 91.33C}$$

$$\Psi_{total} = -61C + 79C + 36.33C + 48.33C + 13.67C + 91.33C$$

$$\boxed{\Psi_{total} = 207.66C}$$

• DELL OPERATOR (∇) / VECTOR OPERATOR / DIFFERENTIAL OPERATOR

Dell operator is mathematical operator used to find gradient, divergent and Curve.

Mathematically the dell operator is given as

$$\nabla = \frac{\partial i}{\partial x} + \frac{\partial j}{\partial y} + \frac{\partial k}{\partial z}$$

Where $\frac{\partial}{\partial x} \rightarrow$ Partial change wrt 'x' in i direction

$\frac{\partial}{\partial y} \rightarrow$ Partial change wrt 'y' in j direction

$\frac{\partial}{\partial x} \rightarrow$ partial change wrt 'z' in k direction

Let f is a Scalar function

The Dell operation on the Scalar function gives

$$\nabla f = \frac{\partial f}{\partial x} i + \frac{\partial f}{\partial y} j + \frac{\partial f}{\partial z} k.$$

→ Applications of Dell operator

1. Application of dell operator on Scalar function gives gradient

$$\nabla \psi = \text{Gradient [Vector]}$$

$$\therefore \nabla \psi = \frac{\partial \psi}{\partial x} i + \frac{\partial \psi}{\partial y} j + \frac{\partial \psi}{\partial z} k$$

2. Application of dell operator on Vector function

Let $A = A_x a_x + A_y a_y + A_z a_z$ is a Vector function

then, $\nabla . A =$ Divergence. [Scalar]

$\nabla \times A =$ Curl. [Vector]

> Summary:

i) $\nabla \varphi$ = Gradient

ii) $\nabla \cdot A$ = Divergence.

iii) $\nabla \times A$ = Curl.

• Dell operator in all 3 Co-ordinate Systems.

1. In Cartesian Co-ordinate System, The Co-ordinates are P(x, y, z). Therefore in CS is given by,

$$\boxed{\nabla = \frac{\partial}{\partial x} a_x + \frac{\partial}{\partial y} a_y + \frac{\partial}{\partial z} a_z \Big/ m}$$

2. $$\boxed{\nabla = \frac{\partial}{\partial \rho} a_\rho + \frac{1}{\rho}\frac{\partial}{\partial \phi} a_\phi + \frac{\partial}{\partial z} a_z \Big/ m}$$

The Cylindrical Co-ordinates of System are ρ, ϕ, z. The Dell operator in CS is given as above.

3. The Dell operator in Spherical Co-ordinate System, the Co-ordinates are r, θ, ϕ

$$\boxed{\nabla = \frac{\partial}{\partial r} a_r + \frac{1}{r}\frac{\partial}{\partial \theta} a_\theta + \frac{1}{r\sin\theta}\frac{\partial}{\partial \phi} a_\phi \Big/ m}$$

→ Concept of Divergence. [Derive the expression for Divergence using gauss law]

Consider a flux density D given by

$D = D_x a_x + D_y a_y + D_z a_z$ [Vector]

$$\oint D \cdot ds = Q$$

$$\psi_{total} = \left[\frac{\partial}{\partial x} D_x + \frac{\partial}{\partial y} D_y + \frac{\partial}{\partial z} D_z\right] \Delta x \, \Delta y \, \Delta z$$

According to Gauss law,

where $\Delta x, \Delta y, \Delta z$ is called closed value Δv.

$$\therefore \oint D \cdot ds = Q = \left[\frac{\partial}{\partial x} D_x + \frac{\partial}{\partial y} D_y + \frac{\partial}{\partial z} D_z\right] \Delta v \longrightarrow (1)$$

From equation ①, we can write closed loop integration

$$\frac{\oint D.ds}{\Delta V} = Q = \left[\frac{\partial}{\partial x}D_x + \frac{\partial}{\partial y}D_y + \frac{\partial}{\partial z}D_z\right] \rightarrow ②$$

$$= \frac{Q}{\Delta V}\ C/m^3 \rightarrow ③$$

When the value shrinks to zero

The equation (2) can be re-written as

$$\lim_{\Delta V \to 0} \frac{\oint D.ds}{\Delta V} = \left[\frac{\partial}{\partial x}D_x + \frac{\partial}{\partial y}D_y + \frac{\partial}{\partial z}D_z\right]$$

Divergence D is given by

$$\therefore \boxed{\text{Div } D = \lim_{\Delta V \to 0} \frac{\oint D.ds}{\Delta V}\ C/m^3}$$

- **Divergence (Div D)**

Let D is flux density

$D = D_x a_x + D_y a_y + D_z a_z$

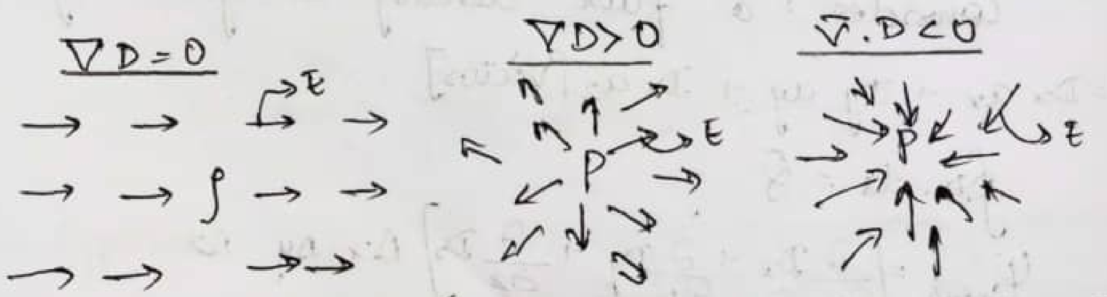
$\nabla . D$ = Divergence of flux density

$\nabla . D > 0$ i.e $\nabla D = +ve$

$\nabla . D < 0$ i.e $\nabla . D = -ve$

$\nabla . D = 0$

The divergence of vector flux density 'D' at a given point is a measure of how much field represented by D diverges or converges from that point.



→ Divergence in all 3 Co-ordinate System.

1. Cartesian Co-ordinate System. $P(x,y,z)$

$P(x,y,z)$

$$D = D_x a_x + D_y a_y + D_z a_z$$

$$\boxed{\nabla \cdot D = \frac{\partial}{\partial x} D_x + \frac{\partial}{\partial y} D_y + \frac{\partial}{\partial z} D_z \ C/m^3}$$

2. Cylindrical Co-ordinate System $P(\rho, \phi, z)$.

$$D = D_\rho a_\rho + D_\phi a_\phi + D_z a_z$$

$$\boxed{\nabla \cdot D = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho D_\rho) + \frac{1}{\rho} \frac{\partial}{\partial \phi} D_\phi + \frac{\partial D_z}{\partial z} \ C/m^3}$$

3. Spherical Co-ordinate System $P(r, \theta, \phi)$

$$D = D_r a_r + D_\theta a_\theta + D_\phi a_\phi.$$

$$\boxed{\nabla \cdot D = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 D_r) + \frac{1}{r \sin\theta} \frac{\partial}{\partial \theta} (\sin\theta \, D_\theta) + \frac{1}{r \sin\theta} \frac{\partial D_\phi}{\partial \phi} \ C/m^3}$$

***

• The Maxwell's 1st equation in Electrostatic

(or)

Gauss Law in point form.

$$\nabla \cdot D = \frac{\partial}{\partial x} D_x + \frac{\partial}{\partial y} D_y + \frac{\partial}{\partial z} D_z = \lim_{\Delta V \to 0} \frac{\oint D \cdot ds}{\Delta V} \longrightarrow (1)$$

According Gauss law,

$$\oint D \cdot ds = Q \ c \longrightarrow (2).$$

For per Unit Volume, the close loop integration of $\oint \frac{D \cdot ds}{\Delta V} = \frac{Q}{\Delta V} c \longrightarrow (3)$

As the volume shrinks to zero, the expression ③ can be re-written as,

$$\lim_{\Delta V \to 0} \frac{\oint D \cdot ds}{\Delta V} = \lim_{\Delta V \to 0} \frac{Q}{\Delta V} \longrightarrow ④$$

Substitute eq ④ in ①

$$\nabla \cdot D = \frac{\partial}{\partial x} D_x + \frac{\partial}{\partial y} D_y + \frac{\partial}{\partial z} D_z = \lim_{\Delta V \to 0} \frac{Q}{\Delta V} \longrightarrow ⑤$$

Since $\lim_{\Delta V \to 0} \frac{Q}{\Delta V} = \rho_v \ C/m^3$.

∴ The equation ⑤ can be re-written as,

$$\boxed{\nabla \cdot D = \frac{\partial}{\partial x} D_x + \frac{\partial}{\partial y} D_y + \frac{\partial}{\partial z} D_z = \rho_v \ C/m^3.}$$

It states that "The electric flux per unit volume leaving a vanishingly very small volume unit is exactly equal to the volume charge density ρ_v".

- **Problems:**

1. Given $D = z \sin\phi \, a_\rho + \rho \sin\phi \, a_z \ C/m^2$. Compute volume charge density ρ_v at a point $P(1, 30, 2)$. ~~The Dete~~

Sol $\nabla \cdot D = \rho_v$

$$\nabla \cdot D = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho D_\rho) + \frac{1}{\rho} \frac{\partial}{\partial \phi} D_\phi + \frac{\partial D_z}{\partial z}$$

$$\nabla \cdot D = \frac{1}{\rho} \frac{\partial}{\partial \rho} [\rho z \sin\phi \ \cancel{a_\rho}] + \cancel{\frac{1}{\rho} \frac{\partial}{\partial \phi} (0)} + \cancel{\frac{\partial}{\partial z} (\rho \sin\phi)}$$

$$= \frac{1}{\rho} z \sin\phi \ \cancel{+ \rho \sin\phi}$$

$$\nabla \cdot D = \frac{1}{1} \times 2 \sin 30^\circ = \underline{1 \ C/m^3}$$

2. Calculate the divergence of the Vector D at the points specified using Cartesian, Cylindrical and Spherical Co-ordinate, where

$D = \frac{1}{z^2}\left[10xyz\,a_x + 5x^2 z\,a_y + (2z^3 - 5x^2 y)a_z\right] C/m^2$. at $(2,3,5)$

$D = 5z^2 a_\rho + 10\rho z\, a_z$ at $P(3, -45°, 5)$,

$D = 2r \sin\theta \sin\phi\, a_r + r\cos\theta \sin\phi\, a_\theta + r\cos\phi\, a_\phi\ C/m^2$.

at $P(3, -45°, -45°)$

Sol $D = \frac{1}{z^2}\left[\overbrace{10xyz\,a_x}^{Dx} + \overbrace{5x^2 z\,a_y}^{Dy} + \overbrace{(2z^3 - 5x^2 y\, a_z)}^{Dz}\right]$.

$$\nabla \cdot D = \frac{\partial}{\partial x}\frac{1}{z^2}10xyz + \frac{\partial}{\partial y}\frac{5x^2 z}{z} + \frac{\partial}{\partial z}\frac{1}{z^2}2z^3 - 5x^2 y.$$

$$= \frac{\partial}{\partial x}\frac{10xy}{z} + \frac{\partial}{\partial y}\frac{15x^2}{z} + \frac{\partial}{\partial z}\frac{2z^3 - 5x^2 y}{z^2}.$$

$$= \frac{10y}{z} + 0 + \frac{z^2(6z^2 - 0) - (2z^3 - 5x^2 y)(2z)}{z^4}$$

$$= \frac{10y}{z} + \frac{6z^4 - [4z^4 - 10x^2 yz]}{z^4}$$

$$= \frac{10yz^3 + 6z^4 - 4z^4 + 10x^2 yz}{z^4}$$

$$= \frac{10(3)(5)^3 + 6(5)^4 - 4(5)^4 + 10(2)^2(3)(5)}{(5)^4}$$

$$\boxed{\nabla \cdot D = 8.96\ C/m^3}$$

$D = 5z^2 a_\rho + 10\rho z\, a_z$

$$\nabla \cdot D = \frac{1}{\rho}\frac{\partial}{\partial \rho}(\rho D_\rho) + \frac{1}{\rho}\frac{\partial}{\partial \phi}D_\phi + \frac{\partial D_z}{\partial z}$$

$$= \frac{1}{\rho}\frac{\partial}{\partial \rho}\rho\, 5z^2 + \frac{1}{\rho}\frac{\partial}{\partial \phi}(0) + \frac{\partial}{\partial z}(10\rho z).$$

$$= \frac{1}{\rho}5z^2 + 10\rho = \frac{1}{3}5(5)^2 + 10(3) = \underline{71.66\ C/m^3}$$

$$D = 2r\sin\theta\sin\phi\, a_r + r\cos\theta\sin\phi\, a_\theta + r\cos\phi\, a_\phi$$

$$r = 3, \quad \theta = -45^\circ, \quad \phi = -45$$

$$\nabla \cdot D = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 D_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\, D_\theta) + \frac{1}{r\sin\theta}\frac{\partial D_\phi}{\partial\phi}\ C/m^3$$

$$= \frac{1}{r^2}\frac{\partial}{\partial r}(r^2\, 2r\sin\theta\sin\phi) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\, r\cos\theta\sin\phi)$$

$$+ \frac{1}{r\sin\theta}\frac{\partial}{\partial\phi}(r\cos\phi)$$

$$= \frac{1}{r^2}[2r\, 2r\sin\theta\sin\phi) + \frac{1}{r\sin\theta}(\cos\theta(r)(-\sin\theta)\sin\phi)$$

$$+ \frac{1}{r\sin\theta} r(-\sin\phi)$$

$$= \frac{1}{3^2}[2(3)2(3)\sin(-45^\circ)\sin(-45^\circ) + \frac{1}{3\sin(45^\circ)}[\cos(-45^\circ)(3)$$

$$(-\sin(-45^\circ))\sin(-45^\circ)] + \frac{1}{3\sin(45^\circ)}(3)(-\sin(45^\circ))]$$

$$\boxed{\nabla \cdot D = 2.73\ C/m^3}$$

3. A Cube of Side 2mtr is centred at origin with ~~adjust~~ edges parallel to Co-ordinate axis of rectangular Co-ordinate System. If $D = \frac{10x^3}{3} a_x$ c/m². find volume charge density also find total charge enclosed by the cube

Sol $D = \frac{10x^3}{3} a_x$ c/m³

$\nabla \cdot D = \rho_v$ c/m³

$Q_{enclosed} = \oint D \cdot ds = \Psi_{total}$.

$$\nabla \cdot D = \frac{\partial}{\partial x} D_x + \cancel{\frac{\partial}{\partial y} D_y}^{0} + \cancel{\frac{\partial}{\partial z} D_z}^{0}.$$

$$\nabla \cdot D = \frac{\partial}{\partial x}\left[\frac{10x^3}{3}\right]$$

$$\nabla \cdot D = \frac{10}{3} \frac{\partial (x^3)}{\partial x}$$

$$= \frac{10}{3} 3x^2$$

$$\boxed{\nabla \cdot D = 10x^2}$$

∴ The volume charge density is equal to

$$D = \int_V \nabla \cdot D \, dv$$

$$= \int_V 10x^2 \, dx \, dy \, dz$$

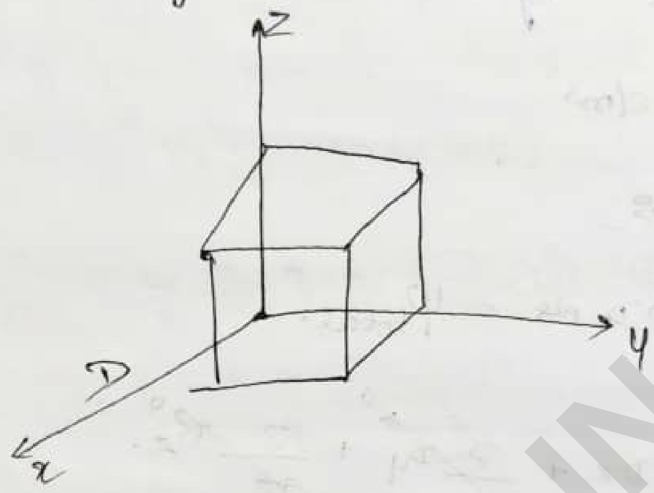
$$= 10 \int_{-1}^{1} x^2 dx \int_{-1}^{1} dy \int_{-1}^{1} dz$$

$$= \frac{10}{3} [1+1] \, [1+1] \, [1+1]$$

$$= \frac{10}{3}(2)(2)(2) = 80/3 \, C$$

$$\boxed{\int \nabla\cdot D\, dv = \frac{80}{3}\, C}$$

$$Q_{enclosed} = \oint D\cdot ds = \Psi_{total}.$$



$$\Psi_{front} = \oint D\cdot ds$$

$$= \oint \frac{10x^3}{3}\, a_x \, dy\, dz\, a_x$$

$$= \frac{10}{3} \oint_{-1}^{1} x^3 \oint_{-1}^{1} dy \oint_{-1}^{1} dz$$

$$= \frac{10}{3} \cancel{\left[\frac{x^4}{4}\right]_{-1}^{1}} [y]_{-1}^{1} [z]_{-1}^{1}$$

$$= \frac{10x^3}{\cancel{12}\,3} [\cancel{2}] [2] [2]$$

$$= \cancel{\frac{10(4)}{12}}\, x^3$$

$$= \frac{10\, x^3}{3}$$

$$= \frac{10x^3}{3} \times (4)$$

$$= \frac{40}{3} x^3 \qquad x = 1$$

$$\boxed{\Psi_{front} = \frac{40}{3}\, C}$$

$$\Psi_{Back} = \oint D \cdot ds$$

$$= \oint \frac{10x^3}{3} dx \; dy \; dz \; (-a_x)$$

$$= -\frac{10x^3}{3} [y]_{-1}^{1} [z]_{-1}^{1}$$

$$= -\frac{10x^3}{3} [2][2]$$

$$= -\frac{10}{3} (-1)^3 (2)(2)$$

$$= \frac{10}{3} \times 4$$

$$\boxed{\Psi_{Back} = \frac{40}{3} C}$$

$$\Psi_{total} = \Psi_{front} + \Psi_{Back}$$

$$= \frac{40}{3} C + \frac{40}{3} C$$

$$\boxed{\Psi_{total} = \frac{80}{3} C/m^3}$$

Note:

$$\int_V \nabla \cdot D = \oint D \cdot ds$$

× (~~A vector field is given by, $A = 30 e^{-r} a_r$~~ ~~$- 2z \, a_z$ C/m². verify divergence theorem~~) ×

4. A Vector field is given by $A = 30e^{-r}a_r - a_z$ a_z C/m². Verify divergence theorem for the volume enclosed by $r = 2$, $z = 0$, $z = 5$.

(or)

Given $D = 30e^{-r}a_r - 2z\,a_z$ C/m². Evaluate Both the sides of divergence theorem for volume enclosed by $r = 2$, $z = 0$, $z = 5$.

Sol $\nabla \cdot D = \frac{1}{r}\frac{\partial}{\partial r} r D_r + \cancel{\frac{1}{r}\frac{\partial}{\partial \phi} D_\phi} + \frac{\partial D_z}{\partial z}$

$$\cancel{\nabla \cdot D = \frac{1}{r}\frac{\partial}{\partial r}(r)(30e^{-r}) + \frac{\partial}{\partial z}2z}$$

$$\cancel{= \frac{1}{r}\frac{\partial}{\partial r}(r)(30(-r)e^{-r-1}) + (2)}$$

$$\cancel{= \frac{1}{2}(r)(30)(-2)(e^{-2-1}) + 2}$$

$$\cancel{= \frac{1}{2}(30)(-2)(e^{-3}) + 2}$$

$$\cancel{= 0.506}$$

$$\nabla \cdot D = \frac{1}{r}\frac{\partial}{\partial r} r(30e^{-r}) + \frac{\partial}{\partial z}(-2z)$$

$$\boxed{\frac{\partial}{\partial x} x e^{-x} = e^{-x} - x e^{-x}}$$

$$\cancel{= \frac{30}{r}\left[e^{-r} - r e^{-r}\right] + 2.}$$

$$\cancel{= \frac{30}{2}\left[e^{-2} - (2)e^{-2}\right] - 2}$$

$$\cancel{= \frac{1}{r}\frac{\partial}{\partial r}}$$

$$= \frac{30}{r}\left[r(-1)e^{-r} + e^{-r}\right] + (-2)$$

$$= \frac{-30}{r}\, re^{-r} + \frac{30}{r}e^{-r} - 2$$

$$= \oint -30e^{-r}\left[1 - \frac{1}{r}\right] - 2 \; dV$$

$$= \int \left(30e^{-r}\left[\frac{1}{r} - 1\right] - 2\right) r\, dr\, d\phi\, dz$$

$$= \int \frac{30e^{-r}}{r}\, r\, dr\, dz\, d\phi - \int 30e^{-r} r\, dr\, d\phi\, dz - 2\int r\, dr\, d\phi\, dz.$$

$$= \left[\int_{r=0}^{2} 30e^{-r}\, dr - \int_{r=0}^{2} 30e^{-r} r\, dr - 2\int_{r=0}^{2} r\, dr\right] \int_{\phi=0}^{2\pi} d\phi \int_0^5 dz$$

$$\cancel{= \left[30\left[-re^{-r-1}\right]_0^2 - 30\left[e^{-r}(1) + r(-re^{-r-1})\right]_0^2 - 2\left(\frac{r^2}{2}\right)_0^2\right] [\phi]_0^{2\pi}[z]_0^5}$$

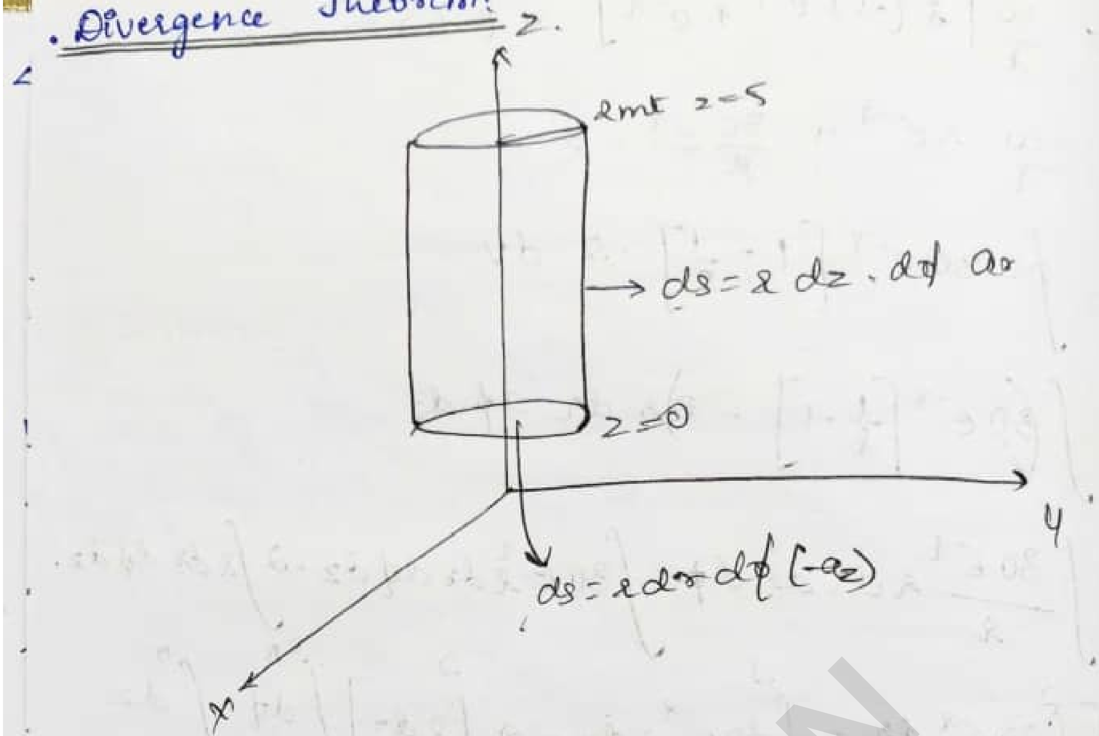
$$\cancel{= 30\left[-2e^{-2-1}\right] - 30\left[e^{-2} + 2(-2e^{-2+1}) - e^0\right] - 2\left[\frac{2^2}{2}\right](2\pi)(5)}$$

$$= 30\left[\frac{e^{-r}}{-1}\right]_0^2 - 30\left[e^{-r}\left(\frac{r^2}{2}\right) - \frac{r^3}{6}(-re^{-r}) + \frac{r^4}{24}\right] \quad (\text{Do in calci}).$$

$$= \left[(25.93) - (17.81) - 2(2)\right](2\pi)(5)$$

$$\boxed{\nabla \cdot D = 129.43\, C}$$

• Divergence Theorem:



5. If $D = \frac{5r^2}{4} a_r \ C/m^2$ in Spherical System then Evaluate both the sides of divergence theorem for volume enclosed by $r = 4m$, $\theta = \frac{\pi}{4}$

Sol: $\int_V \nabla \cdot D = \oint D \cdot ds$.

$$\nabla \cdot D = \frac{1}{r^2}\frac{\partial}{\partial r} r^2 D_r + \cancel{\frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta D_\theta)} \to 0 + \cancel{\frac{1}{r\sin\theta}\frac{\partial}{\partial\phi} D_\phi} \to 0$$

as Spherical $\theta, \phi = 0$

$$= \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{5r^2}{4}\right)$$

$$= \frac{5}{4r^2}\frac{\partial}{\partial r}(r^4)$$

$$\nabla \cdot D = \frac{5}{4r^2}(4r^3)$$

$$\boxed{\nabla \cdot D = 5r \ C/m^3}$$

$$\int_V \nabla\cdot D\, dv = \int_V \nabla\cdot D \times r^2 \sin\theta\, d\theta\cdot d\phi\cdot dr$$

$$= \int_V 5r\, r^2 \sin\theta\, d\theta\cdot d\phi\, dr$$

$$= 5\left[\int_{r=0}^{4m} r^3\, dr \int_{\theta=0}^{\pi/4} \sin\theta\, d\theta \int_{\phi=0}^{2\pi} d\phi\right]$$

$$= 5\left[\frac{r^4}{4}\right]_0^4 \left[-\cos\theta\right]_0^{\pi/4} \left[\phi\right]_0^{2\pi}$$

$$= \frac{5}{4}\left[4^4\right]\left[-\cos\frac{\pi}{4} + \cos 0\right]\left[2\pi\right]$$

$$\boxed{\int_V \nabla\cdot D\, dv = 588.89\, C}$$

$$\int_V \nabla\cdot D = \oint D\cdot ds$$

$$588.89\,C = \oint D\cdot ds$$

$$\oint D\cdot ds = \Psi_{total}$$

For a sphere $\Psi_{total} = \Psi_{\text{outer surface}} + \Psi_{\text{side surface}}$

$$\Psi_{\text{outer surface}} = \oint D\cdot ds$$

$$= \oint \frac{5}{4} r^2 a_r \cdot r^2 \sin\theta\, d\theta\, d\phi\, a_r$$

$$= \frac{5}{4} \cancel{\oint} r^4 \int_0^{\pi/4} \sin\theta \int_0^{2\pi} d\phi$$

$$= \frac{5}{4} r^4 \left[\cos\frac{\pi}{4} - \cos 0\right]\left[2\pi\right]$$

$$= \frac{5}{4} 4^4 \left[\cos\frac{\pi}{4} + \cos 0\right]\left[2\pi\right]$$

$$\boxed{\Psi_{\text{outer surface}} = 588.89\, C}$$

→ Evaluate Both the Sides of divergence theorem if $D = \frac{5}{4} r a_r$ for the Spherical Co-ordinate for the Value enclosed b/w $r = 1mt$ and $r = 2mt$.
[Solved above]

6. Evaluate both the Sides of divergence theorem for the field $D = 2xy\, a_x + x^2 a_y$ C/m² and rectangular parallel pipe formed by the planes $x = 0$ and 1, $y = 0$ and 2, $z = 0$ and 3.

(or)

Verify Both the Sides of divergence theorem if $D = 2xy\, a_x + x^2 a_y$ C/m² present in the region bounded by $0 \le x \le 1$, $0 \le y \le 2$, $0 \le z \le 3$.

Sol $\int \nabla \cdot D = \oint D \cdot ds$

$$\nabla \cdot D = \frac{\partial}{\partial x} D_x + \frac{\partial}{\partial y} D_y + \cancel{\frac{\partial}{\partial z} D_z}^{0}$$

$$= \frac{\partial}{\partial x} 2xy + \cancel{\frac{\partial}{\partial y} x^2}^{0}$$

$$= 2[x(0) + y(1)]$$

$$= \cancel{2x(0)} + 2y$$

$$\boxed{\nabla \cdot D = 2y \ C/m^3}$$

$$\int \nabla \cdot D\, dv = \int 2y \times dx\, dy\, dz$$

$$= 2\left[\int_0^1 dx \int_0^2 y\, dy \int_0^3 dz\right]$$

$$= 2\,[1] \left[\frac{y^2}{2}\right]_0^2 [3]$$

$$= \frac{\cancel{2}(1)}{\cancel{2}}(4)(3)$$

$$\boxed{\int \nabla \cdot D\, dV = 12C}$$

$$\int_V \nabla \cdot D = \oint D \cdot ds$$

$$\oint D \, ds = \psi_{total}.$$

$$\underline{\psi_{total} = \psi_{front} + \psi_{Back} + \psi_{Right} + \psi_{left}}$$

$$\psi_{front} = \oint D \cdot ds$$

$$= ~~\oint \frac{5}{4} r^2 a_r \, dy \, dz \, a_r~~$$

$$= \oint 2xy \, a_x \, dy \, dz \, a_x$$

$$= 2x \oint_0^2 y \, dy \int_0^3 dz$$

$$= 2x \left[\frac{y^2}{2}\right]_0^2 \left[z\right]_0^3$$

$$= (4)(3) x$$

$$= 12x$$

$$= 12(1)$$

$$\boxed{\psi_{front} = 12C}$$

$$\psi_{Back} = \oint D \cdot ds$$

$$= \oint 2xy \, a_x \, dy \, dz \, (-a_x)$$

$$= -2x \oint_0^2 y \, dy \int_0^3 dz$$

$$= -12x$$

$$x = 0$$

$$\boxed{\psi_{Back} = 0}$$

$\psi_{right} = \oint D \cdot ds$

$= \oint x^2 a_y \, dx \, dz \, a_y$

$= \oint_0^1 x^2 dx \oint_0^3 dz$

$= \left(\frac{x^3}{3}\right)_0^1 [z]_0^3$

$= \frac{1}{3}(3)$

$\boxed{\psi_{right} = 1 \, C}$

$\psi_{left} = \oint D \cdot ds$

$= \oint x^2 a_y \, dx \, dz (-a_y)$

$= -\oint_0^1 x^2 dx \oint_0^3 dz$

$= -\left[\frac{x^3}{3}\right]_0^1 [z]_0^3$

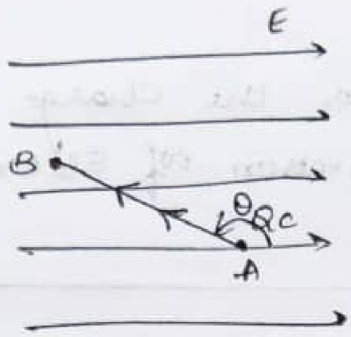
$\boxed{\psi_{left} = -1 \, C}$

$\psi_{total} = 12C + 0C + \cancel{1}C - \cancel{1}C$

$\boxed{\psi_{total} = 12C}$

Verified

Energy expanded in moving a point charge in uniform Electric field.



Consider a point charge Qc at a point A in a uniform electric field E, the force acting on the point charge is given by $\boxed{QE = F}$

$$F = QE \rightarrow ①$$

Let us consider that the charge Q is moving through a distance ΔL along an arbitrary direction A to B which is inclined at an angle of θ to the direction of electric field.

Since the charge is moving opposite to the direction of E, the force acting on the charge is now given as,

$$F = QE\cos(\pi - \theta)$$
$$F = -QE\cos\theta$$

∴ The work-done is given by

$$W = Fd$$
$$W = -EQ\cos\theta \, \Delta L \text{ J.}$$
$$\boxed{W = -QE \cdot \Delta L \text{ J}}$$

If A charge is moved through a differential distance dL in Electric field then differential workdone is given by, $dW = -QE\,dL$ J

∴ The total workdone is given by:

$$\boxed{W = -Q\int_A^B E \cdot dl \ J}$$

- The minus sign indicates the charge is moved opposite to the direction of Electric field.

• **Key note:**

when the point charge is moved ⊥ to the direction of the Electric field the workdone is zero.

$$F = EQ\cos(\pi - \theta)$$

$$\boxed{F = -EQ\cos\theta}$$

Since $\theta = 90°$

$$F = -EQ\cos(90°)$$

$$F = 0$$

$$W = 0$$

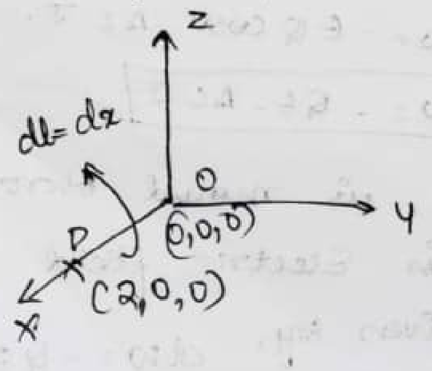
• **Problems:**

1. Given $E = \overset{Dx}{2x\,a_x} - \overset{Dy}{4y\,a_y}$ V/mt. Find the workdone in moving a point charge 2C

i) from (2,0,0) to (0,0,0)

ii) from (0,0,0) to (0,2,0)

Sol



i) $W = -Q\int_A^B E\cdot dl$ J

$= -2\int_2^0 2x\,dx$

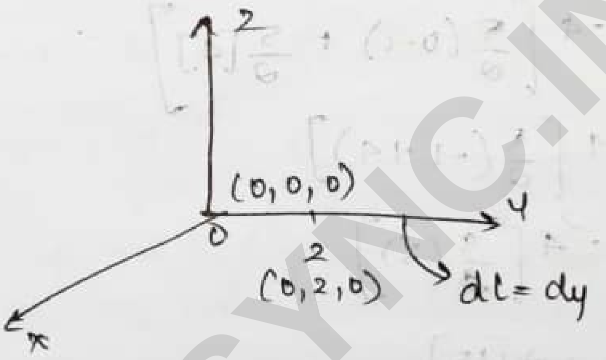
$= -2\left[\frac{\cancel{2}x^2}{\cancel{2}}\right]_2^0$

$= -2[x^2]_2^0$

$= -2[0-(4)]$

$\boxed{W = 8J}$

ii) from (0,0,0) to (0,2,0)



$W = -2\int_0^2 -4y\,dy$

$= 2\left[-\frac{\cancel{4}y^2}{\cancel{2}}\right]_0^2$

$= 4[y^2]_0^2$

$= 4[4]$

$\boxed{W_{ap} = 16J}$

2. Calculate workdone in moving 4 C charge from B(1,0,0) to A(0,2,0) for given electric fields

i) $E = 5x\,a_x + 5y\,a_y$ V/mt.

ii) $E = 5x\,a_x$

iii) $E = 5a_x$

Sol i)

$E = 5x\,a_x + 5y\,a_y$

$W = -Q\int_A^B E \cdot dl$

$W = -4\left[\int_1^0 5x\,dx + \int_0^2 5y\,dy\right]$

$= -4\left[\left[\frac{5x^2}{2}\right]_1^0 + \left[\frac{5y^2}{2}\right]_0^2\right]$

$= -4\left[\frac{5}{2}(0-1) + \frac{5}{2}[4]\right]$

$= -4\left[\frac{5}{2}(-1+4)\right]$

$= -\cancel{4}^{2}\left[\frac{5}{\cancel{2}}(3)\right]$

$= -2[15]$

$\boxed{W = -30\ J}$

ii) $E = 5x\,a_x$

$= -4\int_1^0 5x\,dx$

$= -\cancel{4}^{2}\left[\frac{5x^2}{\cancel{2}}\right]_1^0$

$= -10(-1)$

$\boxed{E = 10\ J}$

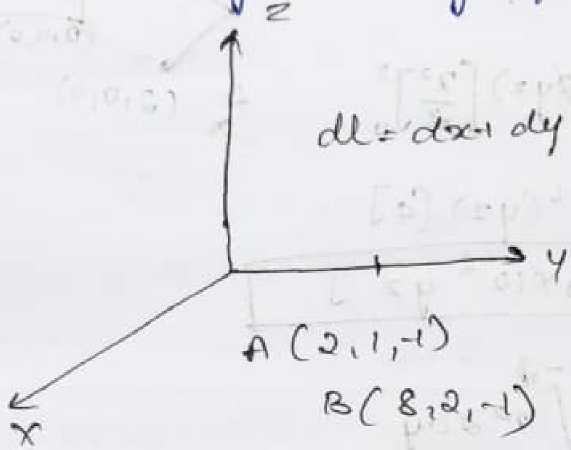
iii) $W = -4\int_1^0 5\,dx$

$W = -4\,[5x]_1^0$

$W = -20\,[0-1] = 20J$ $\boxed{W = 20J}$

3. Determine workdone in carrying a charge of 2C from (2,1,-1) to (8,2,-1) in the given Electric field $E = y\,a_x + x\,a_y$ V/mt.

Sol



If co-ordinate are same then it is zero.

$z = -1$ & $z = -1$ is 0.

$$W = -Q\int_A^B E \cdot dl$$

$$= -2\left[\int_2^8 y\,dx + \int_1^2 x\,dy\right]$$

$$= -2\left[y[x]_2^8 + x[y]_1^2\right]$$

$$= -2[y(6) + x(1)]$$

$$= -2[6y + x]$$

$$= -12y - 2x$$

$$\boxed{W = -2x - 12y}$$

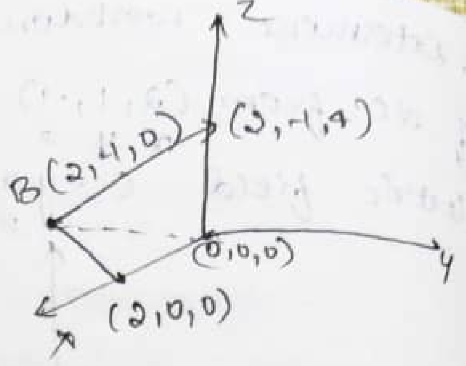
4. Find the work done in moving $5\mu C$ charge from the origin to P(2,-1,4) through an Electric field $E = 2xyz\,a_x + x^2 z\,a_y + x^2 y\,a_z$ V/mt.

i) Straight line segments (0,0,0) to (2,0,0), to (2,-1,0) to (2,-1,4).

Sol. $w_1 = -Q\int_A^B E\cdot dl$

$= -5\times10^{-6}\int_0^2 2xyz\,dx$

$= -5\times10^{-6}(\cancel{2}yz)\left[\frac{x^2}{\cancel{2}}\right]_0^2$



$= -5\times10^{-6}(yz)[4]$

$$\boxed{w_1 = -20\times10^{-6}\,yz\ J}$$

$w_2 = -5\times10^{-6}\int_0^{-1} x^2\,2\,dy$

$= -5\times10^{-6}\times2x^2[y]_0^{-1}$

$= -5\times10^{-6}\times2x^2(-1)$

$$\boxed{w_2 = 10\times10^{-6}\,x^2\ J}$$

$w_3 = -5\times10^{-6}\int_0^4 x^2y$

$= -5\times10^{-6}\,x^2y\,[z]_0^4$

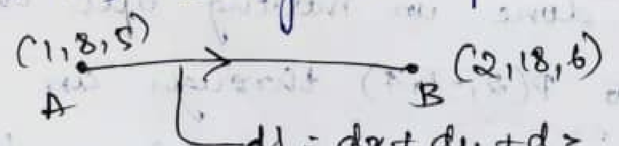
$= -5\times10^{-6}\,x^2y\,[4]$

$$\boxed{w_3 = -20\times10^{-6}\,x^2y\ J}$$

---

5. If $E = -8xy\,a_x - 4x^2a_y + a_z$ V/mt. Find work done in carrying 6C charge from A(1,8,5) to B(2,18,6) along the path $y = 3x+2$; $z = x+4$.

Sol



$w = -6\left[\int_1^2 -8xy\,dx + \int_8^{18} -4x^2dy + \int_5^6 dz\right]$

$w = -6\left[-\overset{4}{\cancel{8}}y\left(\frac{x^2}{\cancel{2}}\right)_1^2 + (-4x^2)\,(y)_8^{18} + (z)_5^6\right]$

$= -6[-4y(4-1) + (-4x^2)(10) + (1)]$

$= -6[-4y(3) + (-4x^2)(10) + 1]$

$= -6[12y - 40x^2 + 1]$

$= 72y + 240x^2 - 6$

$= 72(3x+2) + 240\left(\frac{y-2}{3}\right)^2 - 6$

$W = 216x + 144 + \frac{240}{9}(y^2 + 4 - 4y) - 6$

$W = -6\left[\int -8x(3x+2)dx - 4\left(\frac{y-2}{3}\right)^2 dy + dz\right]$

$= -6\left[\int_{x=1}^{2} -8x(3x+2)dx - \int_{8}^{18} 4\left(\frac{y-2}{3}\right)^2 + \int_{5}^{6} dz\right]$

$= -6\left[\int_{1}^{2} -24x^2 - 16x\, dx - 574.814 + (6-5)\right]$

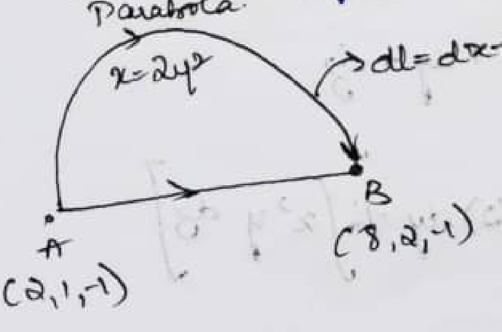
$= -6[-80 - 574.814 + 1]$

$W = 3922.884\,J$

$\boxed{W_{AB} = 3.9\,kJ}$

6. Determine workdone in carrying a charge of $-2C$ from $(2,1,-1)$ to $(8,2,-1)$ in the Electric field given by $E = y a_x + x a_y$. Considering the path along the parabola $x = 2y^2$.

Sol



$x = 2y^2$

$y = \sqrt{\frac{x}{2}}$

$$W_{AB} = -Q\int_A^B E\cdot dl$$

$$= (-2)\left[\int_2^8 y\,dx + \int_1^2 x\,dy\right]$$

$$= 2\left[\int_2^8 \sqrt{\frac{x}{2}}\,dx + \int_1^2 2y^2\,dy\right]$$

$$= 2\left(\frac{28}{3} + \frac{14}{3}\right)$$

$$\boxed{W_{AB} = 28\ J}$$

---

7. Find the work done in moving $5\mu C$ charge from origin to a point $P(2,-1,4)$ through an electric field $E = 2xyz\,a_x + x^2(2)a_y + x^2y\,a_z$ via straight line path $x=-2y$ $z=2x$.

Sol

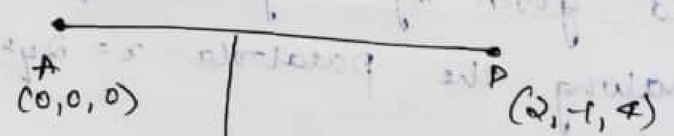
$x = -2y$ $z = 2x$

$y = \frac{x}{2}$ $x = \frac{z}{2}$

$-2y = \frac{z}{2}$

$y = \frac{-z}{4}$

$$\boxed{W = -Q\int E\cdot dl}$$



$dl = dx + dy + dz$

$$W_{AP} = -5\mu\left[\int 2xyz\,dx + \int x^2(2)\,dy + \int x^2y\,dz\right]$$

$$= -5\mu \left[\int_0^2 2(x)\left(\frac{x}{-2}\right)(2x)\,dx + \int_0^{-1} 2(-2y)^2\,dy + \int_0^4 \left(\left(\frac{z}{2}\right)^2 \left(\frac{z}{-4}\right) dz \right) \right]$$

$$W_{AP} = -5\mu \left[-8 + \left(\frac{-8}{3}\right) + (-4) \right]$$

$$\boxed{W_{AP} = 73.33\,\mu J}$$

---

8. Calculate the workdone in moving 4C charge from B(1, 0, 0) to A(0, 2, 0) along the path $y = 2 - 2x$, $z = 0$ in the given electric field $E = 5x\,a_x + 5y\,a_y$ V/m.

Sol

$$W = -Q \int_A^B E \cdot dl$$

$$= -4 \left[\int_1^0 5x\,dx + \int_0^2 5y\,dy \right]$$

$$= -4 \left[\frac{5}{2} + 10 \right]$$

$$\boxed{W = -50\,J}$$

---

9. Given a uniform electric field $E = y\,a_x + x\,a_y + 2a_z$ V/m determine workdone in carrying 2C charge from B(1, 0, 1) to A(0.8, 0.6, 1) along the shorter 'R' of the circle $x^2 + y^2 = 1$, $z = 1$

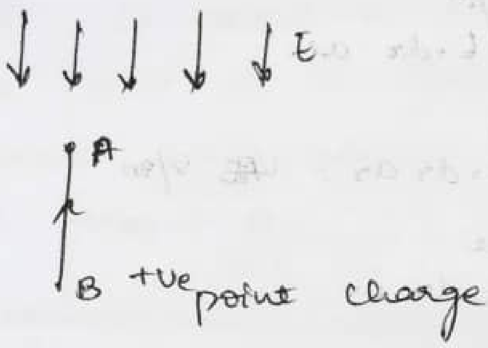
Sol

10. Determine the workdone in carrying 2μC charge from (2,1,-1) to (8,2,-1) in given electric field $E = y a_x + x a_y$ V/m along a hyperbola $x = \frac{8}{7-3y}$

Sol

• POTENTIAL DIFFERENCE AND POTENTIAL.

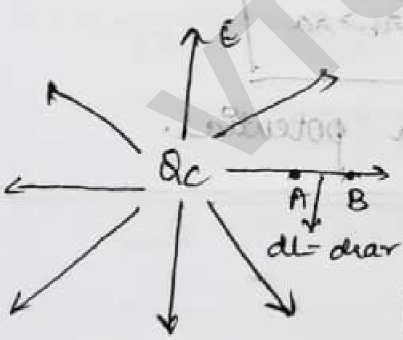
Potential Difference The Potential of a point A with respect to B is defined as workdone in moving a unit positive charge from point A to B against direction of Electric field



$$W = -Q\int_B^A E \cdot dl$$

$$V_{AB} = \frac{W}{Q} = -\int_A^B E\,dl$$

• Potential difference at radial distances r_a and r_b from a point charge.



Consider a point charge Q Columbs which is placed at the origin O and the electric field due to the point charge is along the radial direction.

The Electric field due to a point charge is given by

$$\boxed{E = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}\, a_r \ \text{V/m}}$$

The potential difference between the points A and B is the work done in moving a unit point charge from B to A in the direction against to the electric field.

Therefore,

$$W = -Q \int_B^A E.dl$$

$$W = -Q \int_B^A E.dr\ a_r.$$

$$\frac{W}{Q} = -\int_B^A E.dr\ a_r = V_{AB}\ V/m$$

$$V_{AB} = -\int_{r_b}^{r_a} E.dr.a_r$$

Substitute for E

$$V_{AB} = -\int_{r_b}^{r_a} \frac{Q}{4\pi\varepsilon_0 r^2} a_r dr\ a_r$$

$$V_{AB} = \frac{-Q}{4\pi\varepsilon_0} \int_{r_b}^{r_a} \frac{1}{r^2} dr$$

$$\boxed{V_{AB} = \frac{Q}{4\pi\varepsilon_0}\left[\frac{1}{r_a} - \frac{1}{r_b}\right] \quad r_b > r_a}$$

↳ Scalar potential.

• ABSOLUTE POTENTIAL:

In absolute potential the point B is at infinite distance. Therefor $r_b = \infty$.

Hence

$$\boxed{V_{AB} = \frac{Q}{4\pi\varepsilon_0}\frac{1}{r_a}}$$

• Definition: It is defined as "The work done required to move a point charge of 1c from ∞ point to a specific point along the radial path against the electric field.

$$\therefore \boxed{V_{AB} = \frac{Q}{4\pi\varepsilon_0}\frac{1}{r_a}}$$

1. Given the field $E = 40xy\,a_x + 20x^2 a_y + 2a_z$ V/m Calculate the potential difference between the point P(1,-1,0), Q(2,1,3), where Q = 2C

Sol

$(1,-1,0)$ P → Q $(2,1,3)$

$dl = dx + dy + dz$

$$W_{PQ} = -Q\int E \cdot dl$$

$$= -Q\int 40xy\,a_x + 20x^2 a_y + 2a_z \cdot dx + dy + dz$$

$$W_{PQ} = -Q\int_{(1,-1,0)}^{(2,1,3)} 40xy\,dx + \int 20x^2\,dy + \int 2dz.$$

$= \cancel{+20}$

For a straight line.

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{y - y_1}{x - x_1}$$

$$\frac{1-(-1)}{2-1} = \frac{y-(-1)}{x-1}$$

$$\frac{2}{1} = \frac{y+1}{x-1}$$

$$2x-2 = y+1$$

$$y = 2x-2-1$$

$$\boxed{y = 2x-3}$$

$$2x = y+1+2$$

$$\boxed{x = \frac{y+3}{2}}$$

$$\cancel{W_{PQ} = -2c\left[40y\left[\frac{x^2}{2}\right]_1^2\right.}$$

$$W_{PQ} = -\cancel{2}c\left[\int 40\,x\,(2x-3)\,dx + \int 20\left(\frac{y+3}{2}\right)^2 dy + \int 2\,dz\right.$$

(Do in calci)

$$= -\cancel{2}c\left[\int (80x^2-120x)\,dx + \frac{20}{2}\int (y^2+9+3y)\,dy + \int 2\,dz\right.$$

$$= -1c\left[80\left[\frac{x^3}{3}\right]_1^2 - 120\left[\frac{x^2}{2}\right]_1^3 + \frac{20}{2}\left[\left[\frac{y^3}{3}\right]_{-1}^{1} + (9) + 3\left[\frac{y^2}{2}\right]_0^3\right] + 2\left[z\right]_0^1\right.$$

$$= -1\left[\frac{80}{3}(7) - \frac{120}{2}[4-1] + \frac{20}{2}\left[\left(\frac{1}{3}+\frac{1}{3}\right) + 9 + \frac{3}{2}[9]\right] + 2(3)\right]$$

$$\boxed{W_{PQ} = +106\ V}$$

→ **Imp Q!**

1. Define electric Scalar potential, Derive an expression for the Same.
2. Define potential difference and absolute potential
3. Derive an expression for potential resulting at a point charge placed at the Origion
4. Explain the Concept of work and potential Obtain the expression for potential difference b/w 2 points due to an electric field produced by a point charge.

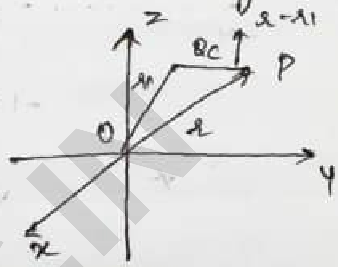
• Potential Due to Several point charges.

The Potential due to a point Charge when located at the Origin is given by.

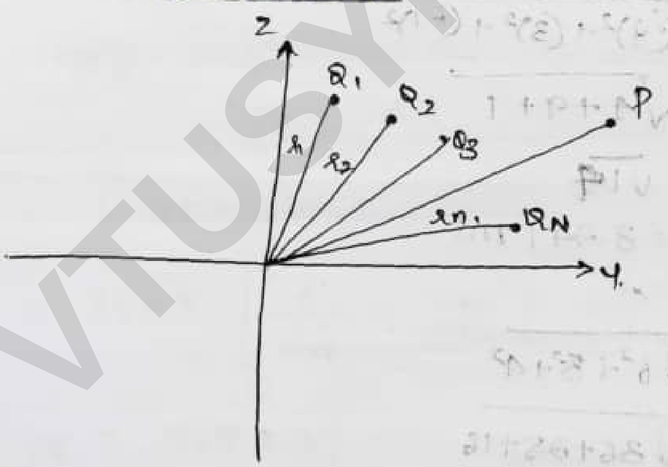
$$V = \frac{Q}{4\pi\varepsilon_0 r}\ V$$

If the point charge is located at Some point other than the Origin, then the potential due to a point charge is given by

$$V = \frac{Q}{4\pi\varepsilon_0|r-r_1|}\ V$$



→ The potential due to Several point charges [Derivation]



$$V_P = V_{PQ_1} + V_{PQ_2} + V_{PQ_3} + \cdots V_{PQ_n}$$

$$= \frac{Q_1}{4\pi\varepsilon_0|r-r_1|} + \frac{Q_2}{4\pi\varepsilon_0|r-r_2|} + \cdots \frac{Q_n}{4\pi\varepsilon_0|r-r_n|}$$

$$= \frac{1}{4\pi\varepsilon_0}\left[\frac{Q_1}{|r-r_1|} + \frac{Q_2}{|r-r_2|} + \cdots \frac{Q_n}{|r-r_n|}\right]$$

$$\boxed{V_P = \frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{n}\frac{Q_i}{|r-r_i|}\ V}$$

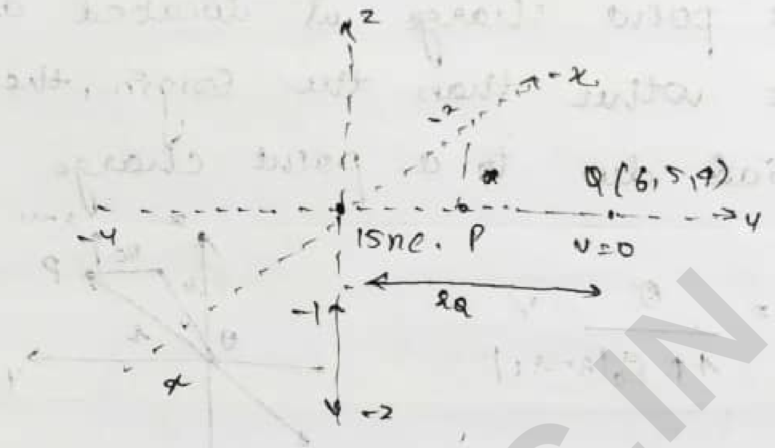
1. A 15 nC point charge is at origin in free space. Calculate V_1 if the point P is located at (-2, 3, -1) and

i) V = 0 at (6, 5, 4)

ii) V = 0 at ∞

iii) V = 5V at (2, 0, 4)

Sol



i) $V_{PQ} = \frac{Q}{4\pi\varepsilon_0}\left[\frac{1}{r_p} - \frac{1}{r_Q}\right]$

$|r_p| = \sqrt{(-2)^2 + (3)^2 + (-1)^2}$

$= \sqrt{4+9+1}$

$= \sqrt{14}$

$|r_p| = 3.741$ mt

$|r_Q| = \sqrt{6^2 + 5^2 + 4^2}$

$= \sqrt{36+25+16}$

$= \sqrt{77}$

$|r_Q| = 8.77$ mt

$V_{PQ} = 15 \times 10^{-9} \times 9 \times 10^{9}\left[\frac{1}{3.741} - \frac{1}{8.77}\right]$

$\boxed{V_{PQ} = 20.69 \text{ Volts}}$

$V_{PQ} = V_P - V_Q$

$= 2269 - 0$

$\boxed{V_{PQ} = 2269\text{ V}}$

ii) $V_{PQ} = \frac{Q}{4\pi\varepsilon_0}\left[\frac{1}{r_p} - \cancel{\frac{1}{r_Q}}\right]^{0} = \frac{Q}{4\pi\varepsilon_0 r_p} = \underline{\underline{36.08\text{ V}}}$

$V_{PQ} = V_P - V_Q$

$= 36.08 - 0$

$\boxed{V_{PQ} = 36.08\text{ V}}$

(iii) $V_{PQ} = \frac{Q}{4\pi\varepsilon_0}\left[\frac{1}{r_p} - \frac{1}{r_Q}\right] =$

$= 9\times10^{9}\times15\times10^{-9}\left(\frac{1}{\sqrt{14}} - \frac{1}{r_Q}\right)$

$|r_Q| = \sqrt{2^2+0^2+4^2}$

$= \sqrt{4+16}$

$|r_Q| = \sqrt{20}$

$= 15\times9\left[\frac{1}{\sqrt{14}} - \frac{1}{\sqrt{20}}\right]$

$\boxed{V_{P} = 5.89\text{ V}}$

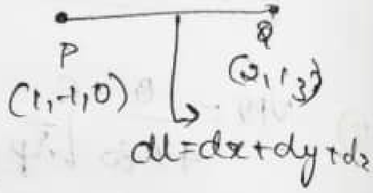
$V_{PQ} = V_P - V_Q$

$= 5.89 - 5$

$\boxed{V_{PQ} = 0.893\text{V}}$

Q. Given field $E = 40xy\, a_x + 20x^2 a_y + 2 a_z$ V/m
Calculate potential between two points
P(1,-1,0) and Q(2,1,3)

Sol: $V_{PQ} = \frac{W}{Q} = -\int E \cdot dl$



$= -\left[\int_1^2 40xy\, dx + \int_{-1}^{1} 20x^2 dy + \int_0^3 2\, dz\right]$

=

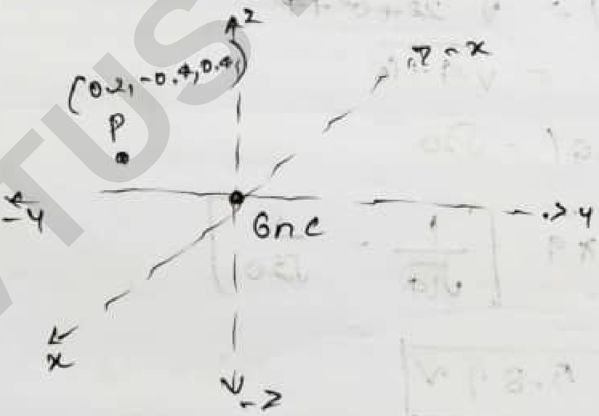
2. A point charge of 6 nC is located at the origin in free space find the potential of point p if p is located at (0.2, -0.4, 0.4) and

i) V=0 at ∞

ii) V=0 at (1,0,0)

iii) V=20 at (-0.5,1,4)

Sol



i) $V_{PQ} = \frac{Q}{4\pi\epsilon_0}\left[\frac{1}{r_P} - \frac{1}{r_Q}\right]$

$|r_P| = \sqrt{0.2^2 + (-0.4)^2 + 0.4^2}$

$r_P = 0.6$

$r_Q =$

$$V_{PQ} = 6\times\cancel{10^{-9}}\times 9\times\cancel{10^{9}}\left[\frac{1}{0.6} - \frac{1}{r_Q}\right]^{10}$$

$$= \frac{6\times 9}{0.6}$$

$$\boxed{V_{PQ} = 90\,Vt}$$

$$V_{PQ} = V_P - V_Q$$

$$= 90V - 0$$

$$\boxed{V_{PQ} = 90V}$$

ii) $$V_{PQ} = \frac{Q}{4\pi\varepsilon_0}\left[\frac{1}{r_P} - \frac{1}{r_Q}\right]$$

$$|r_Q| = \sqrt{1^2+0^2+0^2} = \underline{\underline{1}}$$

$$V_{PQ} = 6\times\cancel{10^{-9}}\times 9\times\cancel{10^{9}}\left[\frac{1}{0.6} - \frac{1}{1}\right]$$

$$= 6\times 9\left[\frac{1}{0.6} - 1\right]$$

$$\boxed{V_{PQ} = 36\,V}$$

$$V_{PQ} = V_P - V_Q$$

$$= 36V - 0$$

$$\boxed{V_{PQ} = 36V}$$

iii) $$V_{PQ} = \frac{Q}{4\pi\varepsilon_0}\left[\frac{1}{r_P} + \frac{1}{r_Q}\right] = 6\times 9\left[\frac{1}{0.6} - \frac{1}{1.5}\right]$$

$$|r_Q| = \sqrt{(0.5)^2+1^2+(-1)^2} \qquad = 54\,Vt$$

$$|r_Q| = 1.5$$

$$\boxed{V_{PQ} = 54\,Vt}$$

$V_{PQ} = V_P - V_Q$

$= 54\,V - 20\,V$

$\boxed{V_{PQ} = 34\,V}$

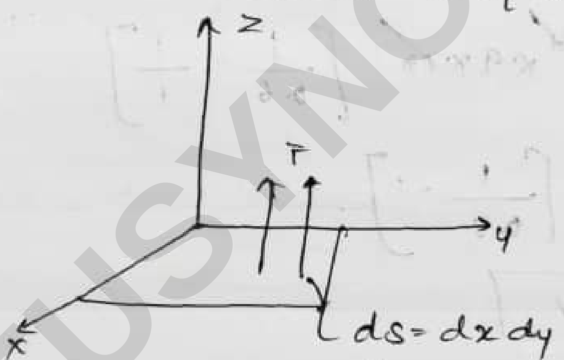
- **Current (I) and Current Density : (J)**

$I \rightarrow$ Scalar

$J \rightarrow$ Vector

Current density [J] is defined as current passing through a unit surface area when the surface area is perpendicular to the direction of the current.

The relation between I & J.



$ds = dx\,dy$

$ds = dx\,dy\,dz \rightarrow$ Vector form

The differential current dI passing through the differential surface ds is given by.

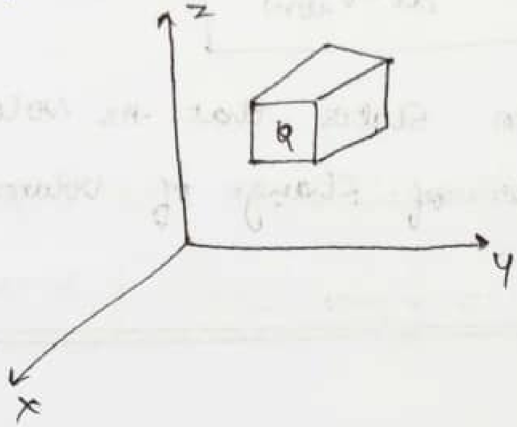
$$dI = J \cdot dS \longrightarrow (1)$$

$$J = \frac{dI}{ds} = \frac{A}{mt^2} \longrightarrow (2)$$

Integrating equation (1) on both sides

$$\boxed{I = \oint J \cdot ds} \longrightarrow (3)$$

* Continuity of Current equation:



The equation proves that as the volume of any closed surface increases there will be no discontinuity of current within the volume.

Consider a closed volume such as cuboid with the charge 'Q' c.

current in any closed surface is given by,

$$I = -\frac{dQ}{dt} \longrightarrow (1).$$

According to relation b/w I & J

$$I = \oint J.ds \longrightarrow (2)$$

from (1) & (2)

$$\oint J.ds = -\frac{dQ}{dt} \longrightarrow (3)$$

Apply Gauss divergence theorem on (3)

$$\oint J.ds = \int_{volume} J.\nabla.dv \longrightarrow (4).$$

Since $Q = \int_{v} \rho_v dv \longrightarrow (5).$

Applying (4) & (5) in (3)

$$\int_{v} J.\nabla.dv = \frac{-d}{dt}\int_{v} \rho_v dv \longrightarrow (6)$$

equn (6) $\rightarrow$ $$\boxed{\nabla \cdot J = -\frac{d}{dt} \rho_{v\,row}}$$

This equation states that As Volume increases w.r.t. 't' the rate of change of Volume charge density decreases.

- **PROBLEMS:**

1. Given vector density, $J = 10\rho^2 z\, a_\rho - 4\rho \cos^2\phi\, a_\phi$ m A/m², find
i) J at $P(\rho = 3, \phi = 30°, z = 2m)$
ii) find I flowing outward through a circle bounded by $\rho = 3mt$, $0 < \phi < 2\pi$, $2 < z < 2.8$ mtr

Sol $J = 10\rho^2 z\, a_\rho - 4\rho \cos^2\phi\, a_\phi$

at P =>

$$J = 10(3)^2(2) a_\rho - 4(3)\cos^2(30°)\, a_\phi \text{ mA/m}^2$$

$$\boxed{J = 180\, a_\rho - 9\, a_\phi \text{ mA/m}^2}$$

Note:
$\rho \rightarrow d s = \rho\, d\phi\, dz\, a_\rho$
$\phi \rightarrow ds = d\rho\, dz(-a_\phi)$

wkt,

$$I = \oint J \cdot ds$$
$$I = I_\rho + I_\phi$$
$$I = \oint J_\rho \cdot ds + \oint J_\phi\, ds \longrightarrow (A)$$

$$I_\rho = \oint J_\rho \cdot ds = \oint 10\rho^2 z\, a_\rho \cdot \rho\, d\phi\, dz\, a_\rho$$
$$= 10\rho^3 \int_0^{2\pi} d\phi \int_0^{2.8} z\, dz$$
$$= 10\rho^3 [2\pi] \left[\frac{(2.8)^2 - 2^2}{2}\right]$$
$$= 10\rho^3 \pi(1.92)$$
$$= 60.2\rho^3$$
$$= 60.288(3)^3 m \qquad \because \rho = 3$$

$$\boxed{I_\rho = 3.257\,A}$$

$31^\circ\ a_y + $

$$I_\phi = \oint J_\phi\, ds$$

$$= \oint -4\rho\cos^2\phi\ a_\phi \cdot d\rho\, dz\,(-a_\phi)$$

$$I_\phi = +4\cos^2\phi \left[\int_0^3 \rho\, d\rho \int_0^{2.5} dz\right]$$

at $\underline{\phi = 0}$

$$= 4\left(\frac{9}{2}\right)\left(\frac{4}{8}\right)$$

$$= 14.4\,mA$$

$$\boxed{I_{\phi=0} = 14.46\,m}$$

$$\boxed{\begin{array}{l} \phi = \phi \Rightarrow d\rho dz\,(-a_\phi) \\ 2\pi \Rightarrow \phi \Rightarrow d\rho dz\, a_\phi \end{array}}$$

at $\underline{\phi = 2\pi}$

$$= \oint -4\rho\cos^2\phi\ a_\phi\, d\rho dz\, a_\phi$$

$$= -4\cos^2(2\pi)\left(\frac{9}{2}\right)\left(\frac{4}{5}\right)$$

$$\boxed{I_{\phi=2\pi} = -14.4\,mA}$$

$$I = I_\rho + I_{\phi=0} + I_{\phi=2\pi}$$

$$= 3.25 + 14.4\times10^{-3} - 14.4\times10^{-3}$$

$$\boxed{I = 3.25\,A}$$

Q. Derive the Continuity of Current. Given $J = -10^6 z^{1.5} a_z\ A/m^2$ in the region where $0 < \rho < 2\mu$ mtr. Find total Current Crossing a Surface $z = 0.1$ mtr.

Sol:

equn (6) -

3. Given Current density $J = \frac{2}{r^2}\cos\theta\, a_r + 20\sin\theta\, a_\theta + r\sin\theta\cos\phi\, a_\phi$ A/m². Find

i) $J = ?$ at $r = 3$ mt., $\theta = 0^\circ$, $\phi = \pi$.

ii) $I_{total} = ?$ passing through a sphere surface $r = 3$ mt. $0 < \theta < 20^\circ$, $0 < \phi < 2\pi$.

Sol

$I = I_r + I_\phi + I_\theta$

$I_r = \oint J_r \cdot ds$

$= \oint \frac{2}{r^2}\cos\theta$

i) $J = \frac{2}{r^2}\cos\theta\, a_r + 20\sin\theta\, a_\theta + r\sin\theta\cos\phi\, a_\phi$ $^{3}1^{\circ}$ $a_y +$

@ $r = 3mt$, $\theta = 0^\circ$, $\phi = \pi$

$J = \frac{2}{9}\cos(0^\circ)a_r + 20\sin(0)a_\theta + (3)\sin(0)\cos(\pi)a_\phi$

$J = \frac{2}{9}a_r + 0 + 0$

$$\boxed{J = \frac{2}{9}a_r \ A/m^2}$$

ii) $I = I_r + I_\phi + I_\theta$.

$I_r = \oint J_r\, ds \longrightarrow r^2\sin\theta\, d\theta\, d\phi$

$= \int \frac{2}{\cancel{r^2}}\cos\theta\, a_r \ \ \cancel{r^2}\sin\theta\, d\theta\, d\phi\, a_r$

$= \int 2\cos\theta\sin\theta\, d\theta\, d\phi.$

$= 2\int_0^{20}\cos\theta\, d\theta \int_0^{20}\sin\theta\, d\theta \int_0^{2\pi} d\phi$

$= 2\left[\sin\theta\right]_0^{20}\left[-\cos\theta\right]_0^{20}\left[2\pi\right]$

$= 2\left[\sin(20) - \sin(0)\right]\left[-\cos(20) + \cos(0)\right]\left[2\pi\right]$

$$\boxed{I_r = 6.79\ A.}$$

$\cancel{I_\phi = \int 20\sin\theta\, a_\theta \ \ \sin\theta\, dr\, d\theta}$

$\cancel{= 20\int_0^{20}\sin^2\theta\, d\theta \int_0^{3} dr.}$

$=$

$$I_\theta = \oint 20 \sin\theta \, a_\theta \cdot \sin\theta \, dr \, d\phi$$

$$= 20 \int \sin^2\theta \, dr \, d\phi$$

$$= 20 \sin^2\theta \int_0^3 dr \int_0^{2\pi} d\phi.$$

$$= 20 \sin^2\theta \, [3] \, [2\pi]$$

$$\boxed{I_\theta = 376.99 \sin^2\theta}$$

4. For a given Current density $J = 4x\, a_x + 3y^2 a_y + 2z^3 a_z$. Find the total Out-coming current from a closed Surface (Cube) Bounded with $1 \le x \le 2$, $2 \le y \le 3$, $3 \le z \le 4$.

Sol

$$I_{top} = \oint J \cdot ds$$

$$= \int 2z^3 a_z \, dx \cdot dy \, a_z.$$

$$= 2\int z^3 \, dx \, dy.$$

$$= 2z^3 \int_1^2 dx \int_2^3 dy$$

$$= 2(4)^3 [2-1][3-2]$$

$$\boxed{I_{top} = 128 \text{ A}}$$

$$I_{Bottom} = \oint J \cdot ds$$

$$= \int 2z^3 a_z \quad dx \, dy (-a_z)$$

$$= -2z^3 \int_1^2 dx \int_2^3 dy$$

$$= (-2)(3)^3 [2-1][3-2]$$

$$\boxed{I_{Bottom} = -54 \text{A}}$$

$$I_{front} = \oint J \cdot ds$$

$$= \oint 4x \; a_x \quad dy \, dz \; a_x$$

$$= 4x \int_2^3 dy \int_3^4 dz$$

$$= (4)(2)(3-2)\,(4-3)$$

$$\boxed{I_{front} = 8 \text{ A}}$$

$$I_{Back} = \oint J \cdot ds$$

$$= \int 4x^2 a_x dy \; dz \; (-a_x)$$

$$= -4x \int_2^3 dy \int_3^4 dz$$

$$= -4(1)(3-2)(4-3)$$

$$\boxed{I_{Back} = -4 \text{ A}}$$

$$I_{right} = \oint J \cdot ds$$

$$= \int 3y^2 a_y \; dx \; dz \; a_y$$

$$= 3y^2 \int_1^2 dx \int_3^4 dz$$

$$= 3(3)^2 [2-1][4-3]$$

$$\boxed{I_{right} = 27 \text{ A}}$$

$$I_{left} = \oint J \cdot ds$$

$$= \oint 3y^2 dy \; dx \; dz (-a_y)$$

$$= -3y^2 \int_1^2 dx \int_3^4 dz$$

$$= (-3)(2)^2 [2-1][4-3]$$

$$\boxed{I_{left} = -12 \text{ A}}$$

$$I_{total} = 128 - 54 + 8 - 4 + 27 - 12$$

$$\boxed{I_{total} = 93 \text{ A}}$$

5. For a given current density by $6xy\, a_x - 3y^2\, a_y$ A/m² and the rectangular path is bounded by $2 \le x \le 5$, $-1 \le y \le 1$, $z = 0$ find the total current I

Sol

$$I_{top} = I_{Bottom} = 0.$$

$$I_{front} = \oint J \cdot ds$$

$$= \oint 6xy\, a_x \; dy\, dz\, a_x$$

$$= 6x \oint_{-1}^{1} y\, dy \oint dz$$

$$= 6x \left[\frac{y^2}{2}\right]_{-1}^{1}$$

$$= \frac{6x}{2}\left[1 - 1\right]$$

$$\boxed{I_{front} = 0}$$

$$I_{Back} = -6x \int_{-1}^{1} y\, dy \int_{0} dz$$

$$\boxed{I_{Back} = 0}$$

$$I_{right} = \oint J \cdot ds$$

$$= \int -3y^2\, a_y \; dx\, dz\, a_y.$$

$$= -3y^2 \int_{2}^{5} dx \cdot \int dz$$

$$= -3(1)^2 \left[5 - 2\right] \cdot \left[0\right]$$

$$\boxed{I_{right} = 0} \quad = \boxed{I_{left}} = 0$$

6. Find the Current Crossing the portion of $x=0$ defined by $\frac{-\pi}{4} \le y \le \frac{\pi}{4}$ and $-0.01 \le z \le 0.01\,m$ If the current density $J = 100\cos(2y)\, a_x\ A/m^2$. Find the Current I Crossing the portion.

Sol

$J = 100\cos(2y)\, a_x\ A/m^2$

$J = J_x a_x + \cancel{J_y a_y}^{\,0} + \cancel{J_z a_z}^{\,0}$

$J = J_\rho a_\rho + J_\phi a_\phi + J_z a_z$

$J = J_r a_r + J_\theta a_\theta + J_\phi a_\phi$.

$ds = dx \cdot dy \cdot dz$

$I = \int J_x \cdot ds$

$= \int 100\cos(2y)\, a_x \cdot dy\, dz (a_x)$

$= 100 \left[\int_{-\pi/4}^{\pi/4} \cos(2y)\, dy \int_{-0.01}^{0.01} dz\ \cancel{a_x \cdot a_x}^{\,1} \right]$

$= 100\,[1]\,[0.01 + 0.01]$

$\boxed{I = 2 \text{ Amperes}}$

$J = 100\cos(2y)\, a_x + 10z^2\, a_z$

$I = \int (J_x + J_z)\, ds$

$= \int J_x \cdot ds + \int J_z\, ds$

$= \int 100\cos(2y)\, a_x \cdot dy\, dz (a_x) + \int 10z^2 a_z\, dz \cdot dy\, a_x$

$= 2\text{ Amp} + 10z^2 \left[\int dx \cdot \int dy\ \cancel{a_z \cdot a_z}^{\,1} \right]$

$= 2\text{ Amp} + 10z^2 \left[0 * \int_{-\pi/4}^{\pi/4} dy \right]$

$= 2\text{Amp} + 0$

$\boxed{I = 2\text{Amp}}$

7. Find for $J = 10^3 \sin\theta \, a_r$ A/m² in Spherical co-ordinate system. Find the current crossing spherical shell with radius $r = 0.02$ mt.

Sol $I = \int J_r \cdot ds$

$= \int 10^3 \sin\theta \, a_r \, r^2 \sin\theta \, d\theta \, d\phi \, a_r$

$= 10^3 r^2 \left[\int_{\theta=0}^{\pi} \sin^2\theta \cdot d\theta \int_{\phi=0}^{2\pi} d\phi \, \cancel{a_r \cdot a_r}^{1} \right]$

$= 10^3 r^2 \left[\int_0^{\pi} \frac{1 - \cos 2\theta}{2} d\theta \int_0^{2\pi} d\phi \right]$

$= 10^3 (0.02)^2 \left[\frac{1}{2}\pi \cdot [2\pi - 0] \right]$

$=$

$\boxed{I = 3.947 \text{ A}}$

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8. Find the current crossing the portion of the plane defined as $y = 0$, $-0.1 \le x \le 0.1$, and $-0.002 \le z \le 0.002$ mt. $J = 10^2 |x| \, a_y$ A/m².

Sol $I = \int J_y \, ds$

$= \int 10^2 |x| \, a_y \cdot dx \, dz \, a_y$

$= 10^2 \int |x| \, dx \, dz \, (1)$

$|x| \to x : x > 0$

$\to -x : x < 0.$

$= 10^2 \left\{ \left[\int_0^{0.1} x \, dx + \int_{-0.1}^{0} (-x) \, dx \right] \int_{-0.002}^{0.002} dz \right\}$

$= 10^2 \left(\frac{1}{200} + \frac{1}{200} \right) (0.002 + 0.002)$

$\boxed{I = 4 \text{ mA}}$

Studied smart, not hard — thanks to VTUSync.in

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